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(54) **INERTIAL NAVIGATION SYSTEM CAPABLE OF DEAD RECKONING IN VEHICLES**

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(71) Applicants: **STMICROELECTRONICS, INC.**,
Coppell, TX (US);
STMICROELECTRONICS S.r.l.,
Agrate Brianza (IT)

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(72) Inventors: **Mahaveer Jain**, Santa Clara, CA (US);
Mahesh Chowdhary, San Jose, CA
(US); **Roberto Mura**, Gallarate (IT);
Nicola Matteo Palella, Rivolta d'Adda
(IT); **Leonardo Colombo**, Lecco (IT)

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(73) Assignees: **STMICROELECTRONICS, INC.**,
Coppell, TX (US);
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Agrate Brianza (IT)

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Primary Examiner — Aniss Chad

Assistant Examiner — Christopher George Fees

(74) *Attorney, Agent, or Firm* — Seed Intellectual
Property Law Group LLP

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(58) **Field of Classification Search**

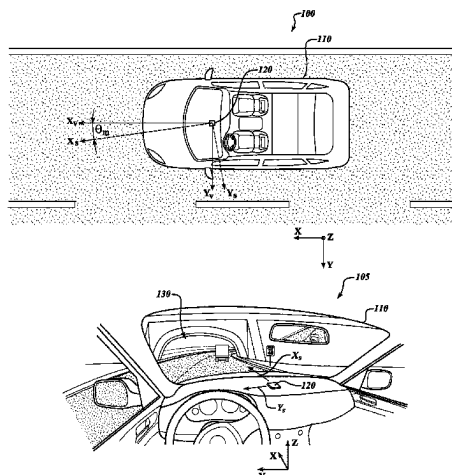
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(57) **ABSTRACT**

A device including microelectromechanical systems (MEMS) sensors are used in dead reckoning in conditions where Global Positioning System (GPS) signals or Global Navigation Satellite System (GNSS) signals are lost. The device is capable of tracking the location of the device after the GPS/GNSS signals are lost by using MEMS sensors such as accelerometers and gyroscopes. By calculating a misalignment angle between a forward axis of a sensor frame of the device and a forward axis of a vehicle frame using the data received from the MEMS sensors, the device can accurately calculate the location of a user or the vehicle of the device even without the GPS/GNSS signals. Accordingly, a device capable of tracking the location of the user riding in the vehicle in GPS/GNSS signals absent environment can be provided.

19 Claims, 12 Drawing Sheets



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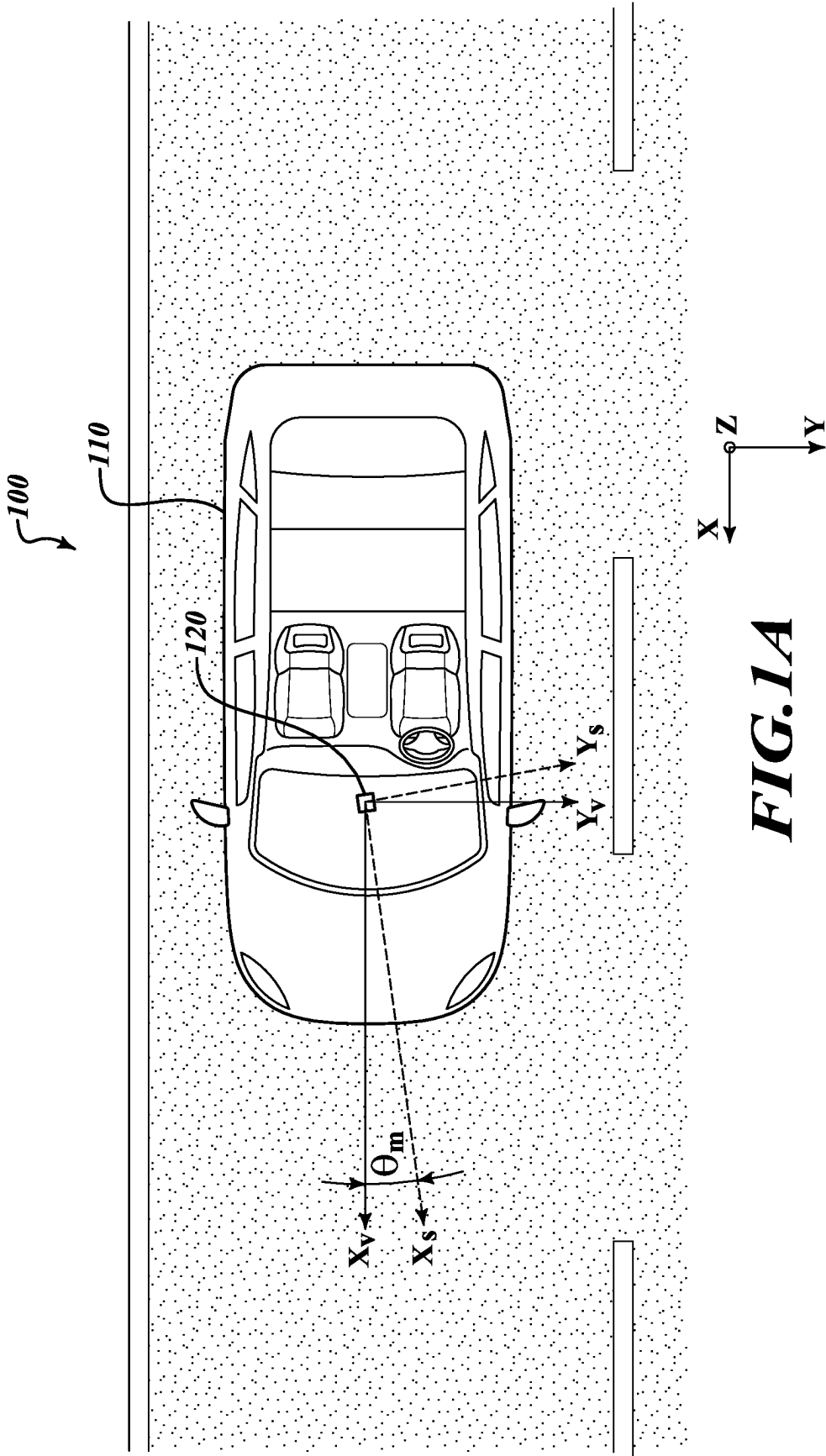
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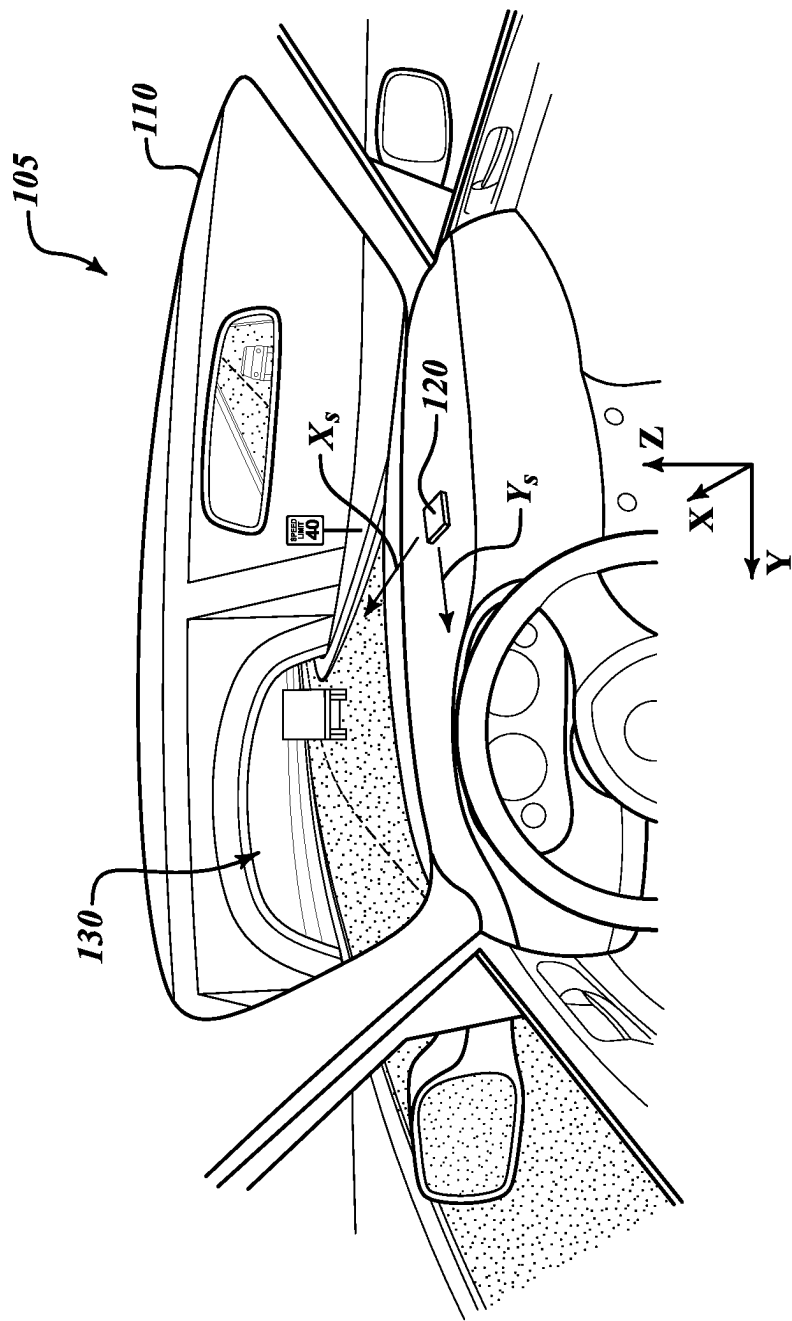
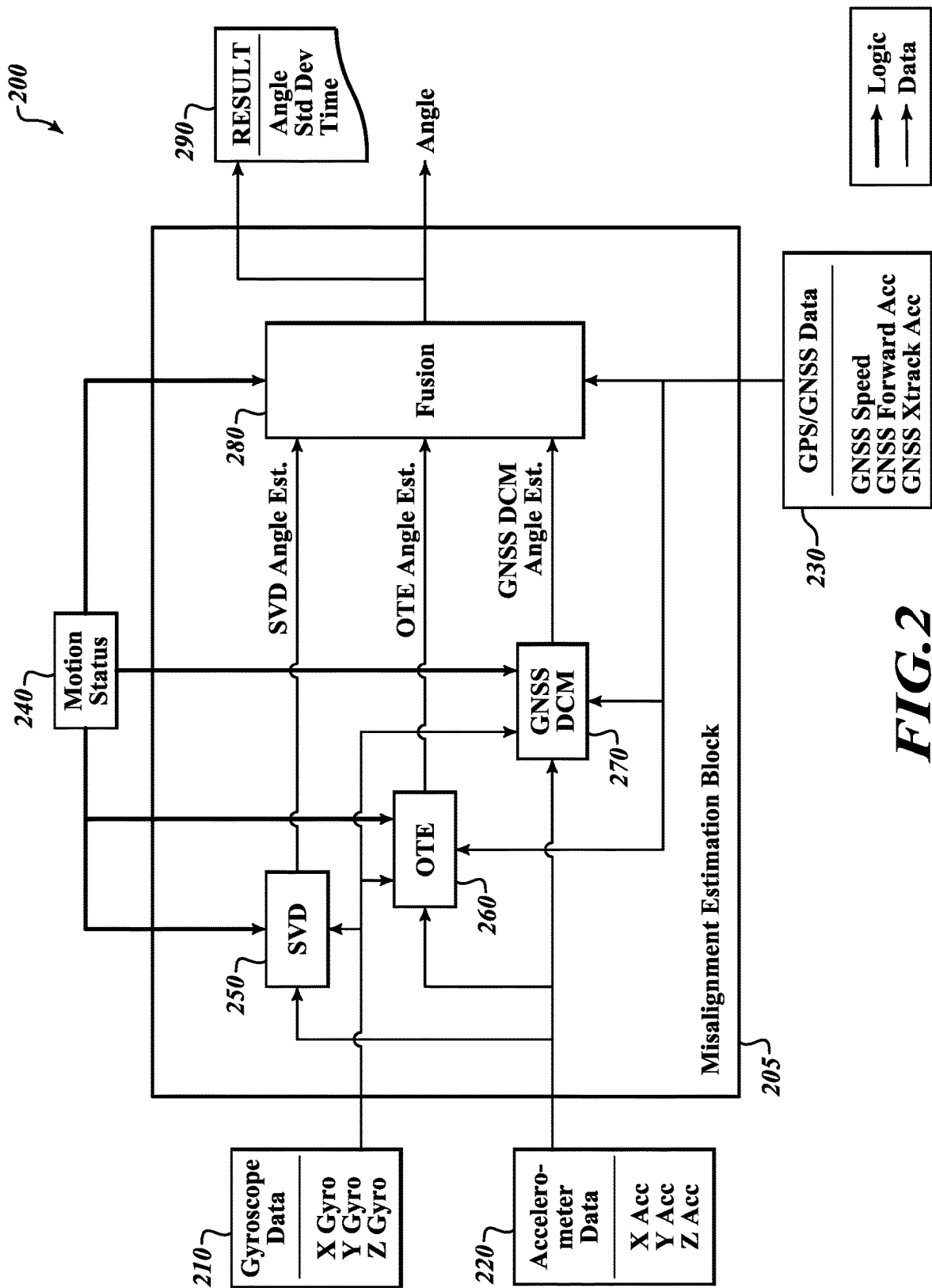
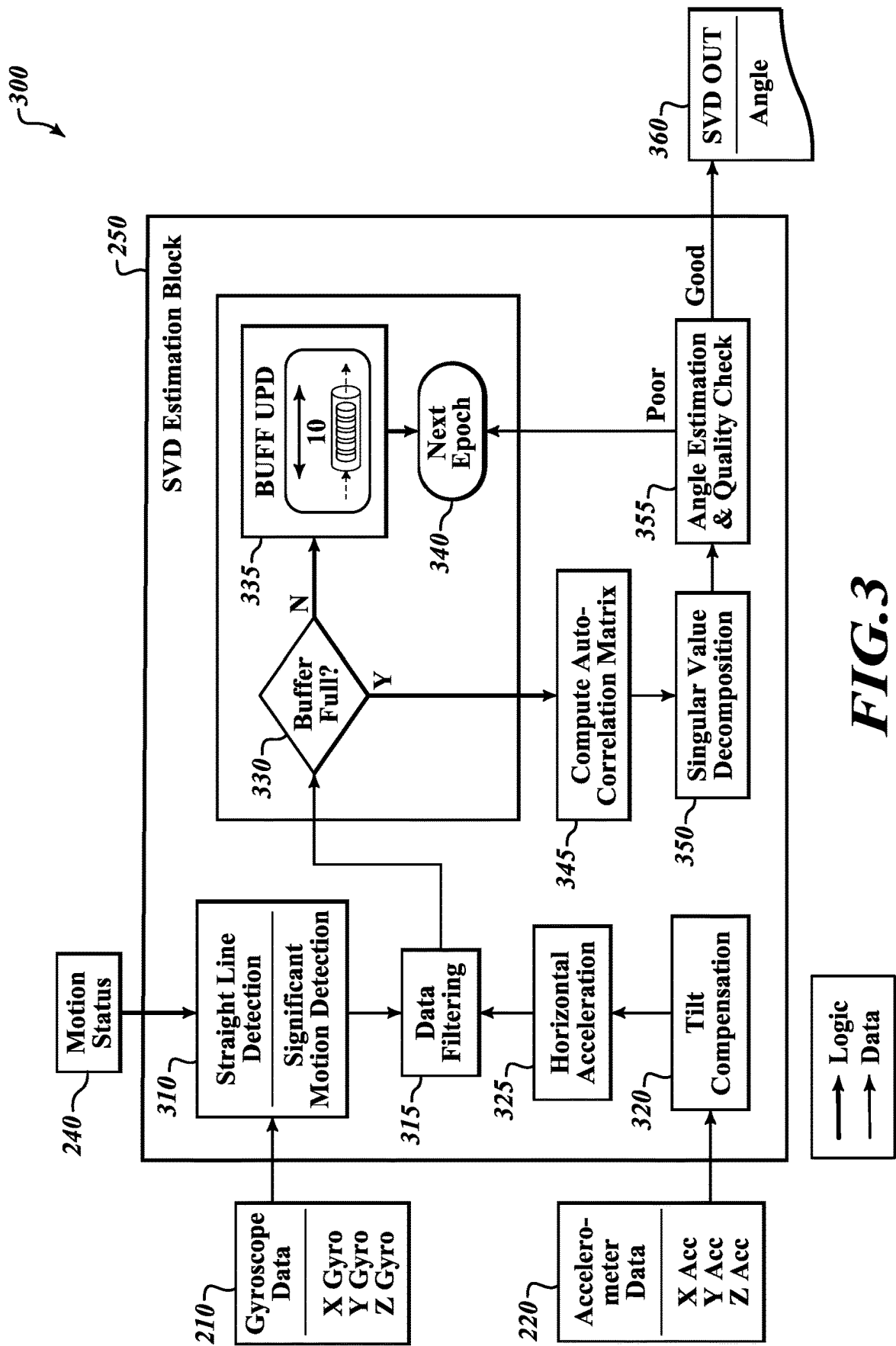


FIG. 1B





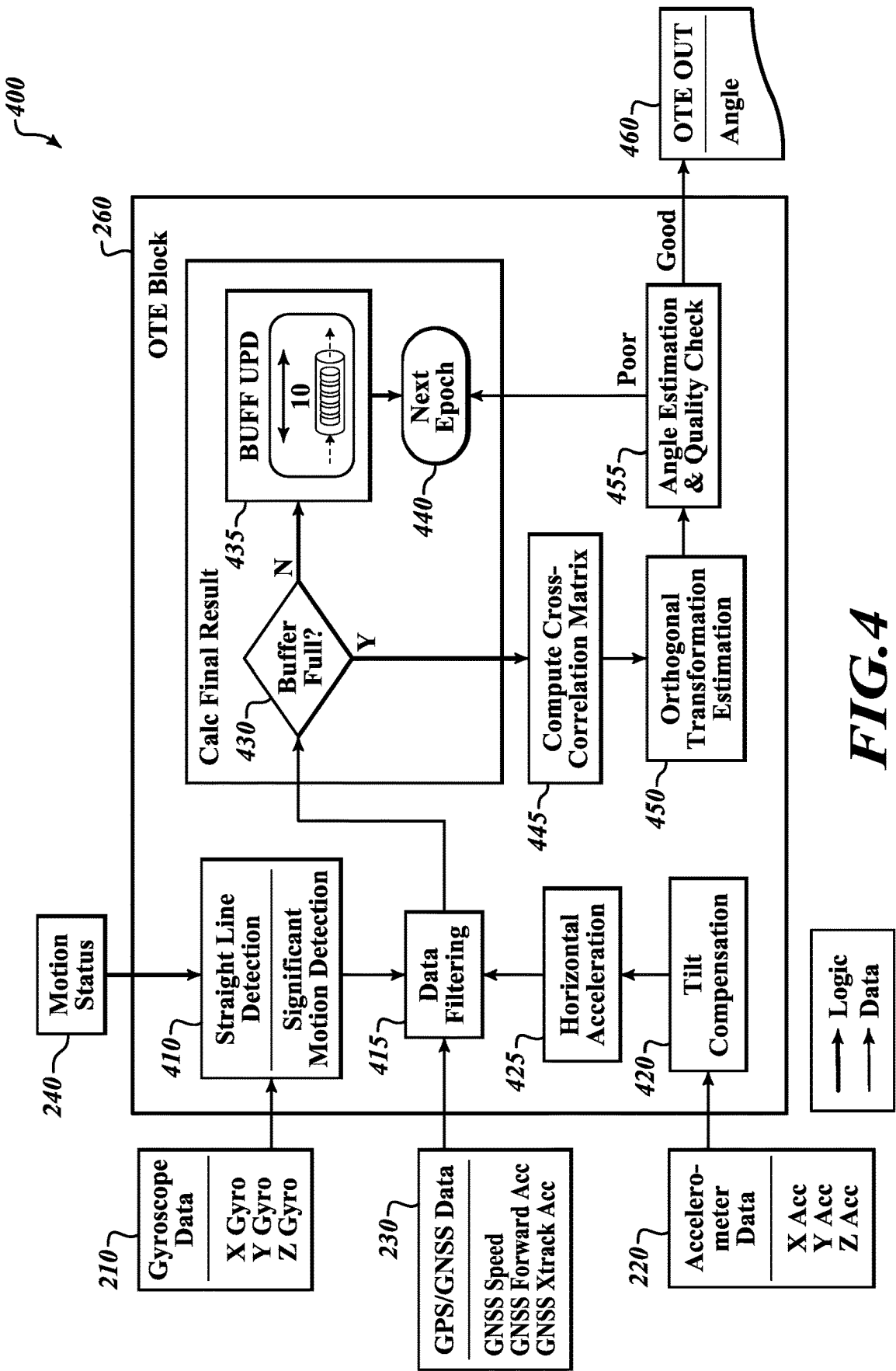


FIG.4

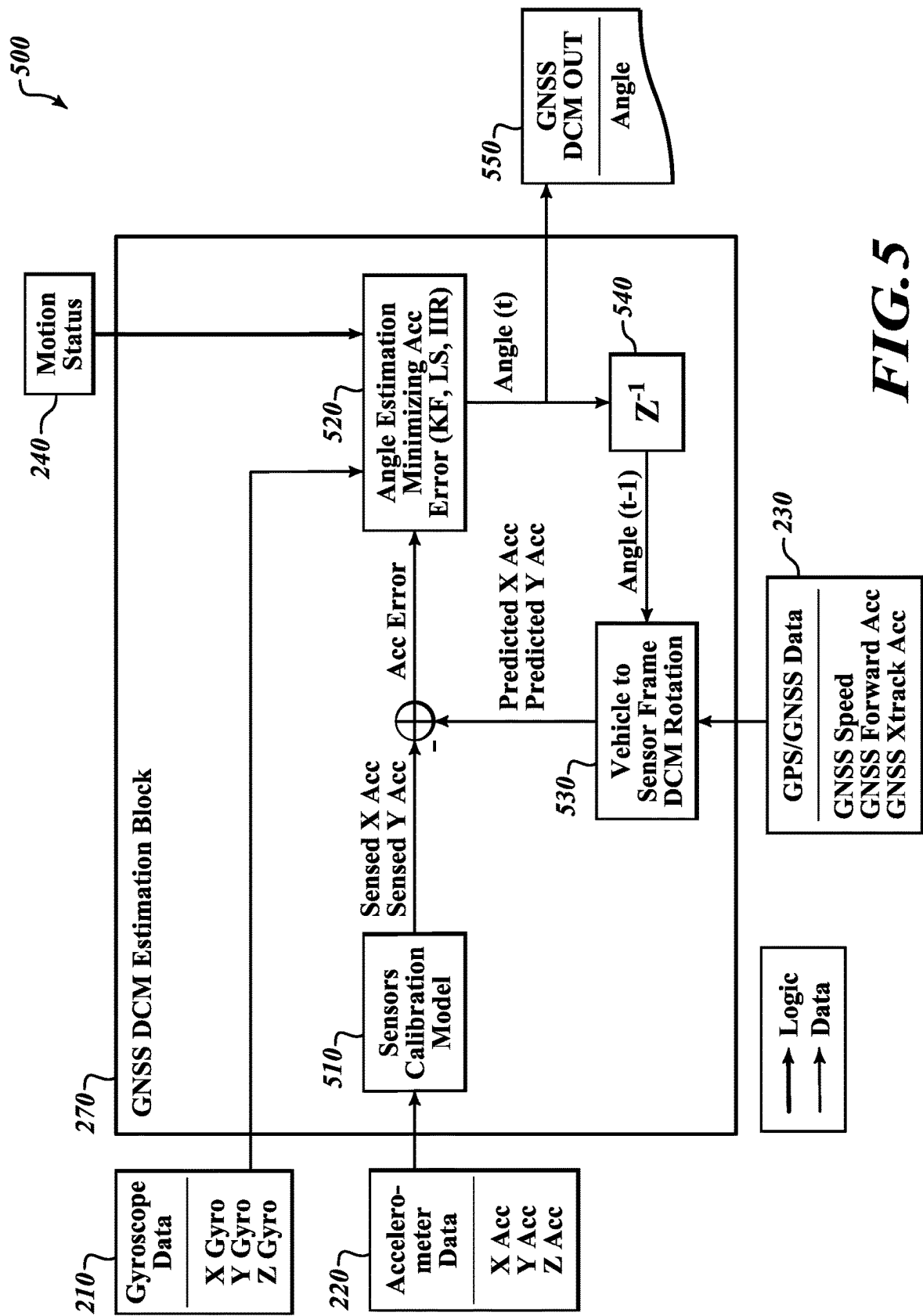


FIG. 5

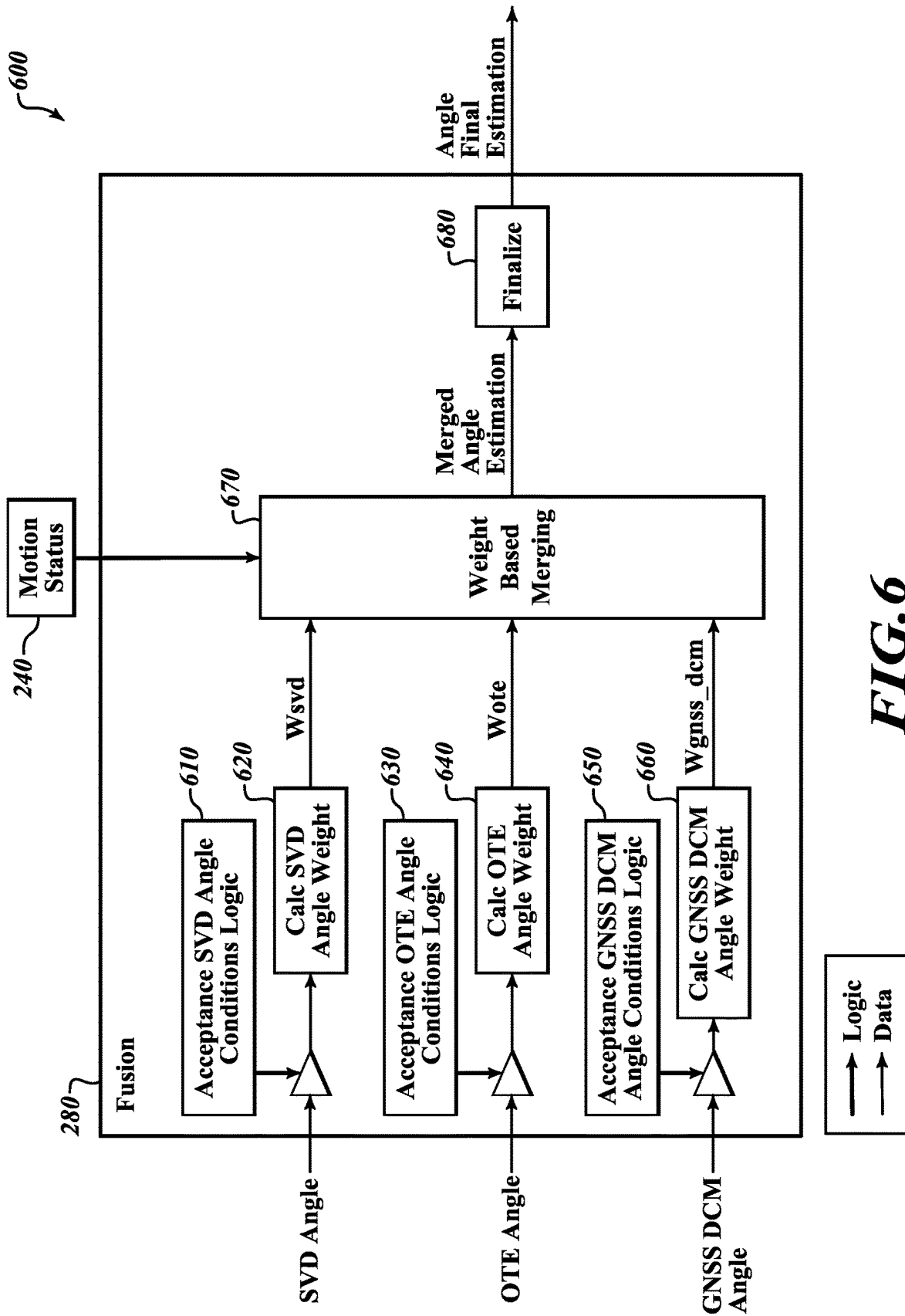
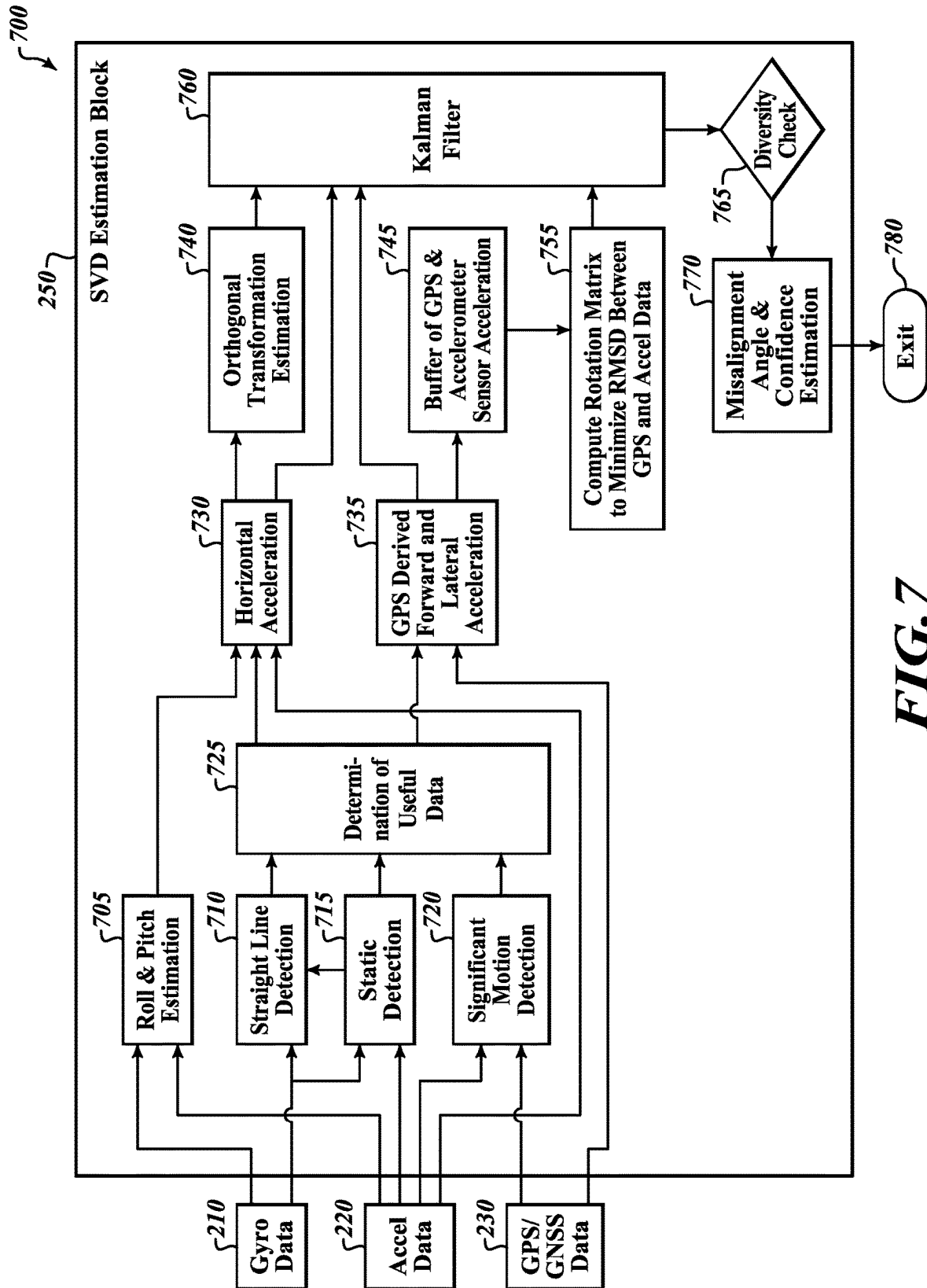


FIG. 6



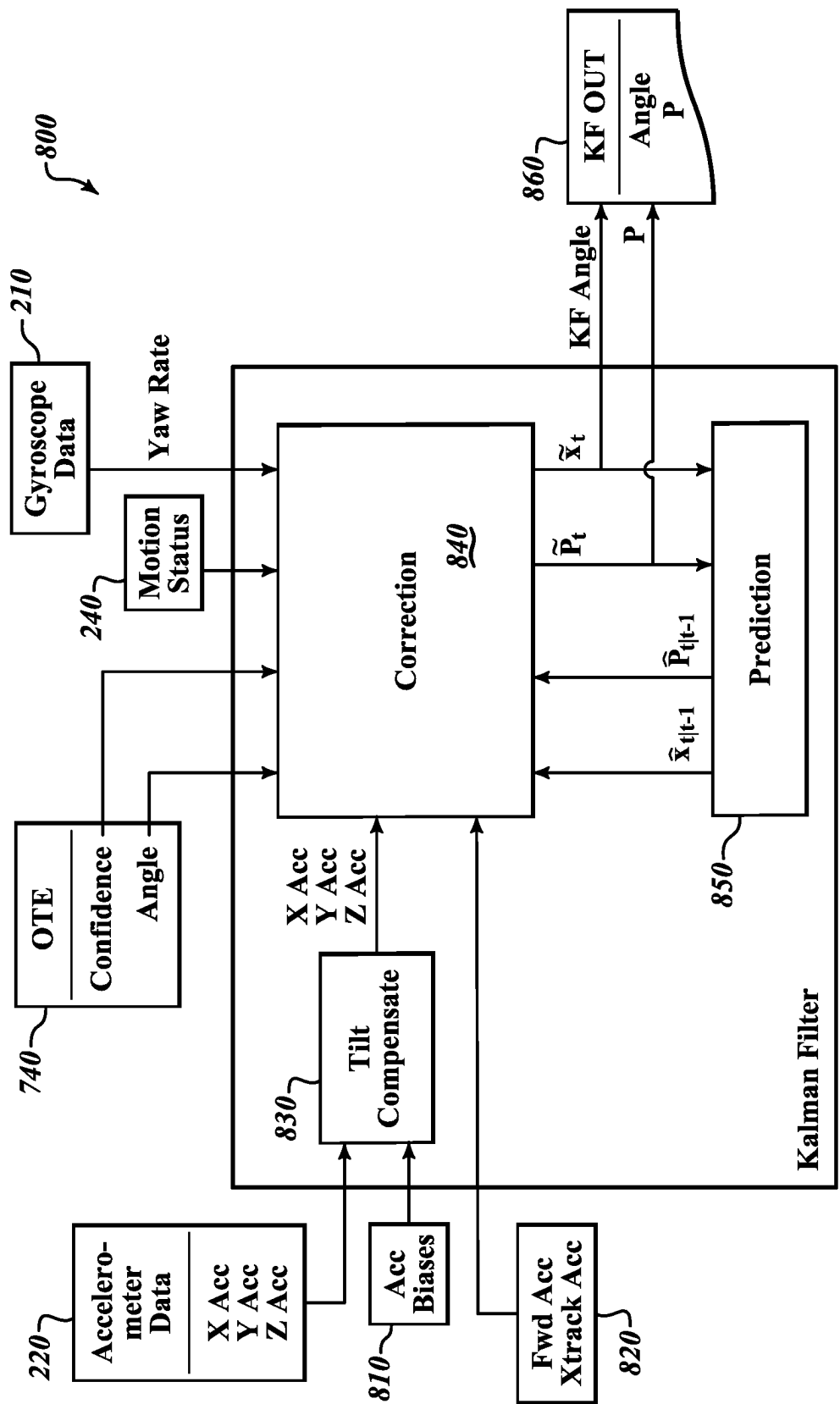
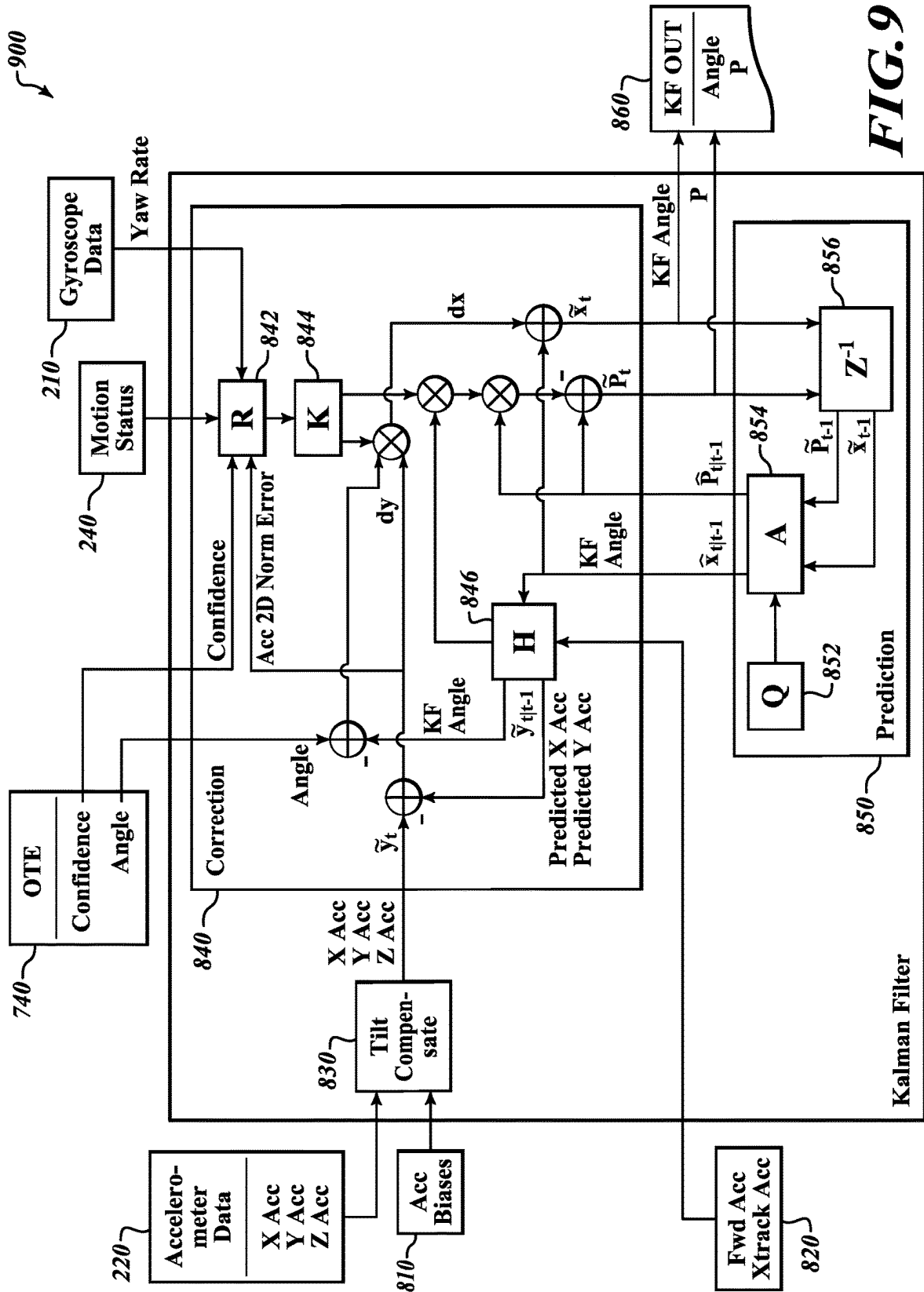
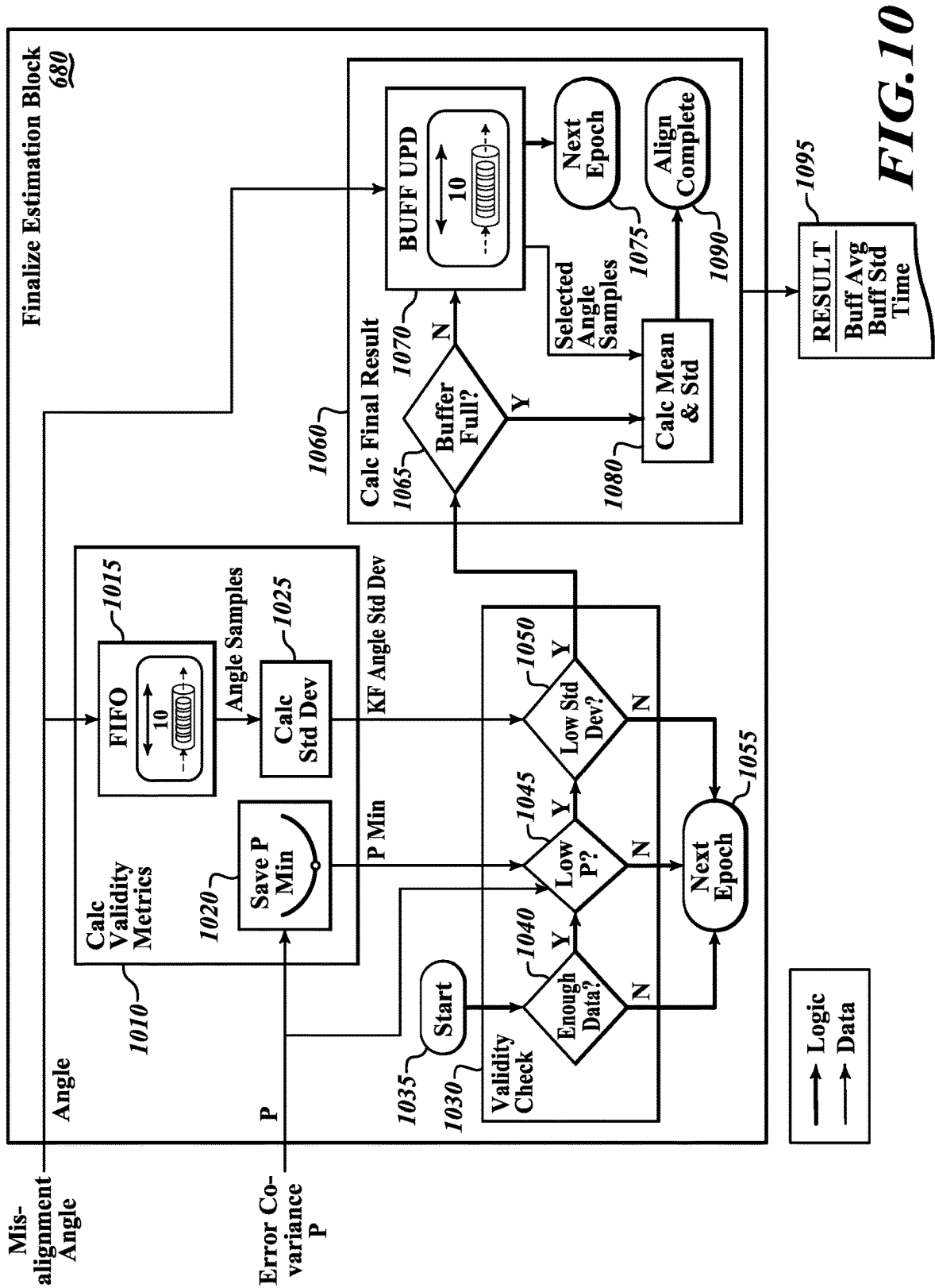


FIG.8





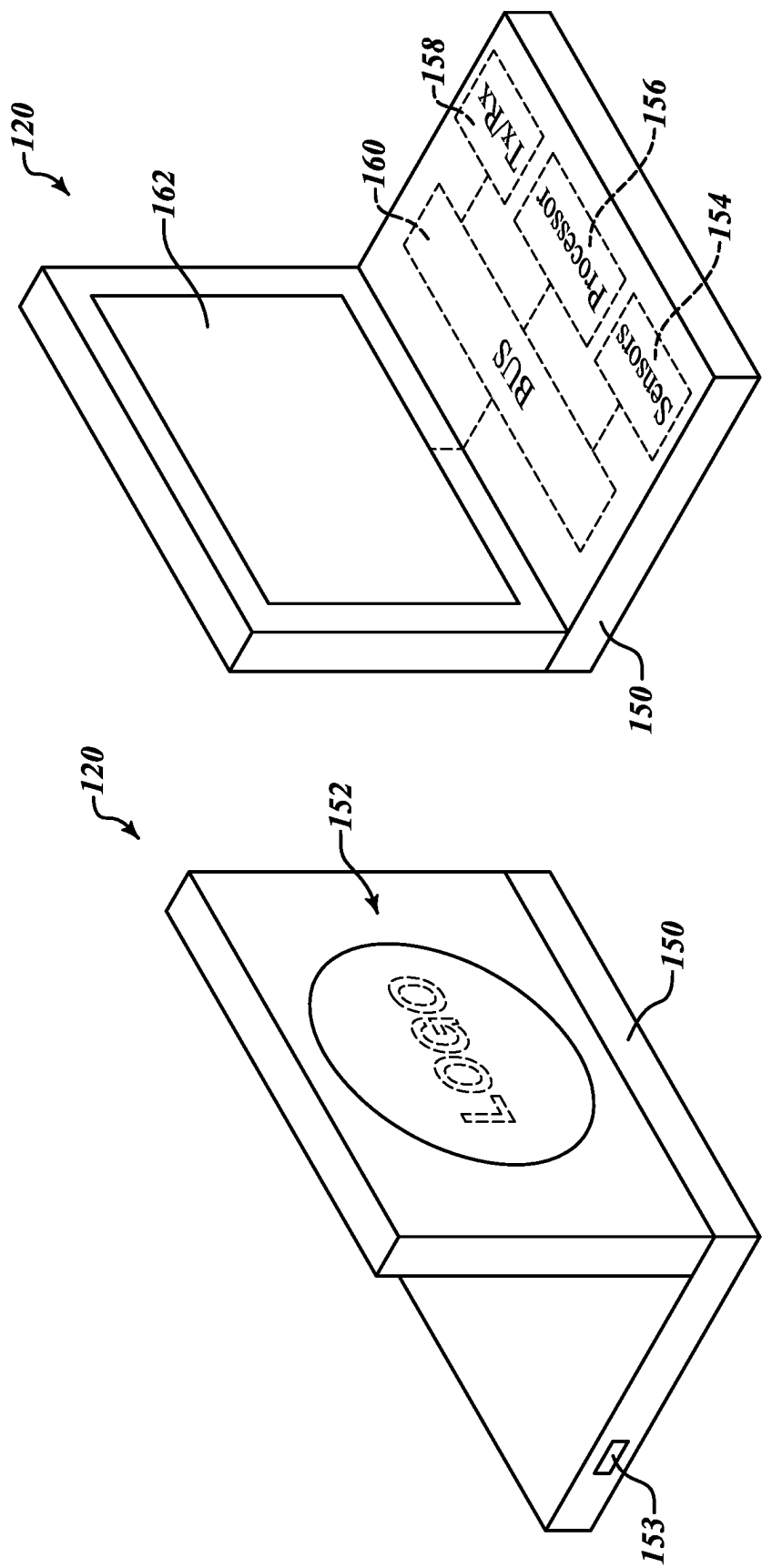


FIG. 11A

FIG. 11B

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INERTIAL NAVIGATION SYSTEM CAPABLE OF DEAD RECKONING IN VEHICLES

BACKGROUND

Technical Field

The present disclosure relates to vehicle dead reckoning using microelectromechanical systems (MEMS) sensors in a stand-alone electronic device.

Description of the Related Art

As electronic devices become more ubiquitous, users' expectations continue to expand. Many users regularly use the location based applications, such as a map application available on current devices, such as cell phones, tablets, navigation devices and infotainment systems installed in vehicles. These location based services are used by both drivers and riders in ride share applications. The drivers use location based services on their mobile devices, sometimes also integrating with location based applications associated with the driver's vehicle. The riders use location based services on their mobile devices from which they are requesting a ride.

The mobile devices and vehicles include a variety of the following components to aid in the location identification of the driver and rider: MEMS sensors, Global Positioning System (GPS) sensors, Global Navigation Satellite System (GNSS) sensors, among others. For example, the MEMS sensor include, among others, accelerometers and gyroscopes. Accelerometers are used in measuring the acceleration forces or rate of change of velocity. Gyroscopes are used in measuring angular velocity. The mobile devices and vehicles, alone or in combination, display the user's position in space on a location based on a display screen. Through these ride share applications, the driver's location can be shared with the rider and the rider's location can be shared with the driver.

However, when either of these users enters into a place where GPS/GNSS signals are weak, disturbed, or unavailable, the location of the user as detected with the device becomes inaccurate or unavailable and can be displayed inaccurately on the other user's device. As such, the device no longer accurately locates the position of the user. In addition, the conventional dead reckoning method of calculating the current position by using a previously determined position (e.g., by using GPS/GNSS sensors) and advancing that position based estimated speeds is not accurate enough.

BRIEF SUMMARY

Due to these short comings in the location based applications and the dead reckoning methods, an improved device and method of accurately analyzing a location of the user is provided, which is especially beneficial when GPS/GNSS signals are not strong enough or unavailable.

The present disclosure provides a method and system to accommodate circumstances when a user is desirous of accurate location information, but does not have a clear path to receive satellite signals, such as GNSS and GPS signals. These dead zones for GPS or GNSS signals are sometimes referred to as urban canyons, where tall buildings and other obstructions prevent consistent or strong signals.

The device includes sensors, a wireless transmitter/receiver, and a processor for gathering, processing, and transmitting data about the device and the environment in which

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the device is located. The device is typically simpler than a mobile phone, such that it can be made cost effectively and bolster the capabilities of the mobile phone or a vehicle when in parking garages, urban canyons, or tunnels.

One application of the present disclosure may be beneficial for ride hailing services, which are services that use online-enabled platforms to connect between passengers (riders) and local drivers using their personal vehicles. Examples of ride hailing services include Uber and Lyft. However, when GPS/GNSS signals are degraded or lost, riders trying to use these ride hailing services lose location of the driver's vehicle making pick up more challenging. By utilizing the device, even in environments where GPS/GNSS signals are lost or weak, the driver of the vehicle, and the rider will more effectively locate the location of the vehicle.

Another application of the present disclosure may be beneficial for analyzing collisions between vehicles. The application can be applied to both autonomous and human-operated vehicles. The device, being removable, but fixed to the vehicle can assist in determining a trajectory of the vehicle based on the sensor alignment methods described. Proper sensor alignment between the device and the vehicle, will give useful information such as impact of crash, direction of impact, etc. when analyzing crashes. The device, with the sensor alignment methods, can provide direction of the related vehicle even in circumstances where the GPS/GNSS signals are degraded or lost. Once a collision has taken place, the aligned sensor data can be used to analyze the collision, including the time right before the collision and impact related data (post collision data).

In particular, the disclosure provides a method for determining a misalignment angle between a vehicle reference frame and a device reference frame, the device being removably attached to the vehicle. The sensors in the electronic device provide data capable of determining the misalignment angle according to the movement of the vehicle using sensors such as accelerometers and gyroscopes.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

For a better understanding of the embodiments, reference will now be made by way of example only to the accompanying drawings. In the drawings, identical reference numbers identify similar elements or acts. The sizes and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes of various elements and angles are not necessarily drawn to scale, and some of these elements may be enlarged and positioned to improve drawing legibility. Further, the particular shapes of the elements as drawn are not necessarily intended to convey any information regarding the actual shape of the particular elements, and may have been solely selected for ease of recognition in the drawings.

FIG. 1A is a top view of a roadway that includes a misalignment angle θ_m between a vehicle and a device positioned in a random, temporarily fixed location in the vehicle according to an exemplary embodiment of the present disclosure;

FIG. 1B is a view of the vehicle entering into a tunnel where there is low or no GPS/GNSS signals capable of being received with the device of FIG. 1A;

FIG. 2 is a system flow diagram for determining the misalignment angle θ_m using the device according to an exemplary embodiment of the present disclosure;

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FIG. 3 is a flow diagram of determining a misalignment angle θ_m under the singular value decomposition (SVD) estimation method according to an exemplary embodiment of the present disclosure;

FIG. 4 is a flow diagram of determining a misalignment angle θ_m under the orthogonal transformation estimation (OTE) method according to an exemplary embodiment of the present disclosure;

FIG. 5 is a flow diagram of determining a misalignment angle θ_m under the GNSS direction cosine matrix (DCM) method according to an exemplary embodiment of the present disclosure;

FIG. 6 is a flow diagram of a fusion block of the outputted misalignment angles according to an exemplary embodiment of the present disclosure;

FIG. 7 is a flow diagram of determining a misalignment angle θ_m under the SVD estimation method according to another embodiment of the present disclosure;

FIG. 8 is a flow diagram of an exemplary Kalman Filter according to one embodiment of the present disclosure;

FIG. 9 is a detailed flow diagram of an exemplary Kalman Filter shown in FIG. 8 according to one embodiment of the present disclosure; and

FIG. 10 is a flow diagram of a finalize estimation block according to another embodiment of the present disclosure.

FIG. 11A and FIG. 11B are front and back perspective views of a location identification device of the present disclosure.

DETAILED DESCRIPTION

In the following description, certain specific details are set forth in order to provide a thorough understanding of various disclosed embodiments. However, one skilled in the relevant art will recognize that embodiments may be practiced without one or more of these specific details, or with other methods, components, materials, etc. In other instances, well-known algorithms or methods associated with dead reckoning have not been shown or described in detail to avoid unnecessarily obscuring descriptions of the embodiments.

Unless the context indicates otherwise, throughout the specification and claims which follow, the word “comprise” and variations thereof, such as, “comprises” and “comprising” are to be construed in an open, inclusive sense that is as “including, but not limited to.” Further, the terms “first,” “second,” and similar indicators of the sequence are to be construed as interchangeable unless the context clearly dictates otherwise.

Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. It should also be noted that the term “or” is generally employed in its broadest sense that is as meaning “and/or” unless the content clearly dictates otherwise.

FIG. 1A is a top view 100 of a vehicle 110 on a roadway. A device 120 is positioned within the vehicle 110. The device

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may be resting in a location in the vehicle, such as on a seat, on the dashboard, or in a center console of the vehicle 110. While the device 120 is not fixed to a certain location with respect to the car, it can be positioned anywhere in or on vehicle 110. In this regard, a relationship of the orientation of the device 120 with respect to the fixed three dimensional Cartesian coordinate system (XYZ axis or vehicle reference frame) of the vehicle may be fluid until it is fixed on a certain location in the vehicle. In some embodiments, the device 120, similar to a conventional, embedded vehicle navigation system, such that the device be provided in a permanently or temporarily fixed manner.

The device 120, for example the device 120 in FIGS. 11A and 11B, can be a standalone device that rests on a dashboard or other temporarily fixed location of the vehicle. This device can be used by drivers of ride share services to increase the accuracy of their location information to be shared with riders and with the service. This device 120 includes a base 150 that may be flexible to conform to and grip to a surface of the dashboard. In other embodiments, this base may include a clip or other attachment to hold the device in place while driving.

The device includes a front panel 152 on which a logo can be displayed. This device includes an illumination device, such as light emitting diodes to illuminate the panel. Alternatively, this panel 152 may be a digital screen such that the displayed image can be changed. For example, during a surge pricing time, this displayed image could be a different color or shape to inform riders of a surge time period. The service could activate all devices 120 in a variety of vehicle within an area with the same surge time period symbol.

The device includes a plurality of sensors 154, a processor 156, a transmitter/receiver device 158, and a bus or data line 160 that couples the plurality of sensors to the processor and the transmitter/receiver 158. The plurality of sensors can include GPS/GNSS sensors, accelerometers, and gyroscopes. The gyroscopes may have a smaller dynamic range than gyroscopes included in mobile devices, such as phones. These sensors are fixed with respect to a body of the device 120. Once positioned in the car, the sensors are then fixed with respect to a vehicle reference frame. The processor can be an application specific integrated circuit or standard data processing chip that includes memory and logic. The transmitter/receiver may include a variety of transmitting and receiving means, such as WiFi™, Bluetooth™, satellite, and other electronic, wireless communication techniques.

In some embodiments, the device 120 includes a display 162, which may be a touch screen display such that the device 120 can be the electronic computing device that provides the ride sharing application interface to the driver, with or without connection to the driver's personal mobile device.

The device includes a power connection 153 in FIG. 11A on a side of the device 120. This can be any type of suitable connector to charge a battery (not shown).

The device has a sensor reference frame that is fixed with respect to the device 120, but can change with respect to the vehicle reference frame as the device is moved to a different position in the vehicle or moved to another vehicle. As such, a misalignment angle θ_m between the vehicle 110 and the device 120 positioned in a random location in the vehicle 110 is calculated in real time, based on the most recent location of the device with respect to the vehicle. For example, the device 120 may be coupled to a holder in the vehicle 110, such as an adjustable arm coupled to the dashboard, the misalignment angle may be calculated based on this location with respect to the vehicle. The system

described in this disclosure will regularly determine if a position of the device has changed since the last check. Different actions are taken depending on whether the position has changed.

The device and methods described in this disclosure provide more accurate location information to identify a location of the device with respect to a map in a location based application. For example, a police officer at the scene of an accident waiting for an ambulance that is traveling through low GPS signal environments will be able to accurately track a position of the ambulance that includes the device 120 as the device 120 can calculate location information from the sensors in the device. The location information can be shared through an emergency department's application, through the cloud, or through other wireless services, like cellular networks from the ambulance to the police officer. The officer's location can also be shared to the ambulance to give each party accurate location information.

In this example arrangement, the vehicle 110 is moving in a first direction. In a navigation perspective, a heading of the vehicle is the direction in which the vehicle is pointing and moving towards, i.e., to the left in the Figure. For example, the first direction may be a heading direction of the vehicle 110 if the vehicle is moving forward. In another example, the first direction may be a tail direction of the vehicle 110 if the vehicle is moving backward. The first direction of the vehicle 110 is indicated as X_v . X_v is a vector that reflects the speed and direction of the vehicle 110. In one or more embodiments, the X_v component reflects not only speed and direction, but also angular velocity and angular acceleration of the vehicle 110.

A second direction that is perpendicular to the first direction may be referred to as a lateral direction Y_v vector. A vehicle 110 may have a lateral component Y_v when the vehicle 110 turns directions (e.g., turning right or left) instead of driving in a straight direction (e.g., driving forward or backward). For example, if the vehicle 110 meets a curved road, the vehicle 110 will both have a forward direction X_v component as well as a lateral direction Y_v component.

Hereinafter, the term "vehicle frame" or "vehicle reference frame" is used to set a direction of the vehicle in relation to the various vectors of the vehicle 110. However, for the purpose of calculating the misalignment angle θ_m , two vector components of the vehicle frame may be considered. For example, the forward direction of the vehicle frame may refer to the X_v vector of the vehicle, and the lateral direction of the vehicle frame may refer to the Y_v vector of the vehicle, which is perpendicular to the X_v vector of the vehicle. The term "sensor frame," will be used in a similar sense to indicate the various vectors of the sensors in the device 120.

In one or more embodiments, the X_v component is measured using an inertial measurement unit (IMU), which can include the plurality of sensors of the device 120. The IMU includes sensors such as a 3-axis accelerometer and a 3-axis gyroscope. The 3 axes accelerometer sensor provides specific forces sensing in three orthogonal directions. The three axes gyroscope sensor provides angular rates sensing in three orthogonal directions. These are sometimes referred to as six Degrees of Freedom (DOF) IMU sensors. The calibrated six DOF IMU sensors have less sensor noise and error and, thus, produce reliable results having increased accuracy for a short duration. By using these sensors, various motion related data may be obtained by the device 120. For example, when the vehicle 110 accelerates or decelerates,

the IMU of the device 120 will detect the acceleration and the angular acceleration of the vehicle 110 and calculate the direction of the force ($F=ma$; m: mass, a: acceleration) which will be indicative of the direction of the vehicle 110.

The IMU can measure and report a device's specific force, speed, movement direction, tilt direction of the device, angular velocity, angular acceleration, etc., using a combination of accelerometers and gyroscopes. The IMU allows analysis of the location of the device when the GPS/GNSS receiver does not work or has a limited signal strength, such as in tunnels, in parking garages, inside buildings, urban canyons, or when electronic interference is present.

In some embodiments, the X_v component may be measured using GPS/GNSS sensors of the device 120. The GPS/GNSS sensors can provide the speed and direction of X_v of the vehicle 110 until the GPS/GNSS signal is lost. The device 120 may utilize the GPS/GNSS information received to analyze and calculate the X_v component. After the signal is lost or the strength of the signal goes below a threshold incapable of locating the device 120, the device 120 may calculate X_v by using the sensors in the IMU. The GPS/GNSS data received until the GPS/GNSS signal is lost may be used in the calculation of the misalignment angle θ_m .

In other embodiments, the X_v component may be measured using an odometer of the vehicle 110. In this embodiment, the vehicle 110 may be connected with the device 120 to communicate or share movement related information such as speed, direction, acceleration rate, angular velocity, angular acceleration, etc. Instead of using the GPS/GNSS sensors, accelerometers, or gyroscopes in the device 120, the device 120 may receive movement information directly from the vehicle 110. The connection between the device 120 and the vehicle 110 may be established through Bluetooth, Wi-Fi™, Zigbee™, NFC™ (Near-field communication) or any suitable wireless connection. The connection can be also established by directly connecting the device 120 to the vehicle 110 by way of wire capable of sending and receiving data. In some embodiments, the battery charging cables for the device 120 that is connected with the vehicle 110 may be utilized to receive motion data from the vehicle 110. The X_v component may also be determined from a combination of these described methods.

The device can be coupled to a mobile device, such as a cellular phone, tablet, or other mobile device that hosts the application of the driver. The device can be paired, or otherwise coupled by a cord or wirelessly to share data between the device and the mobile device of the driver. This can be established through Bluetooth, Wi-Fi™, Zigbee™, NFC™ (Near-field communication) or any suitable wireless connection.

The information or data collected by the device 120 can be processed in the device, in the cloud, or on the mobile device of the driver, such as through the ride share application. The location based information gathered by the device and the sensors once processed can be utilized by the application to display an accurate location of the device with respect to a map of the driver, the rider, and the service.

As mentioned, in one or more embodiments, X_v and Y_v are vectors of the vehicle 110 indicating the speed and direction of the vehicle 110. In other embodiments, the X_v and Y_v vectors of the vehicle 110 may also be indicative of or reflecting any movement related factors suitable for calculating the misalignment angle θ_m such as the acceleration rate, angular velocity, or angular acceleration, etc., of vehicle 110 and the device 120. The X_v and Y_v vectors correspond to the vehicle reference frame. The method for calculating the misalignment angle θ_m between the X_v

vector of the vehicle **110** and the X_S vector of the sensor of the device **120** will be explained in detail throughout the specification.

The IMU which includes accelerometers and gyroscopes is fixed within the device **120** in a certain location or a certain direction so that it helps a user to indicate forward, backward, left and right directions or the north, south, east and west directions of the device **120**, i.e., the sensor reference frame. Similar to the vehicle case, the heading of a device **120** may be determined in a similar manner. The heading of a device **120** or to be exact, the heading of the IMU (or hereinafter “a sensor frame” or “the sensor reference frame”), is the compass direction the sensor frame is pointing towards. In some embodiments, the heading direction of the vehicle frame and the heading of the device **120** are aligned to be in the same direction. If the heading of the vehicle and the heading of the sensor frame are not aligned in the same direction, sensor measurements are not usable for improving the accuracy of position.

In some embodiments, the sensor frame will be capable of finding its own tilt direction of the device **120** within a space using an X, Y, Z coordinate system (Cartesian coordinates) or an r, θ , ϕ spherical coordinate system (r: radial distance, θ : azimuthal angle, ϕ : polar angle) using measurement from the accelerometer and gyroscopes using sensor fusion.

The calculation of the pitch, yaw, and roll of the device **120** assists in bringing the sensor data from the sensor frame, X_S , Y_S to local/vehicle/navigation frame. For the purpose of calculating misalignment angle θ_m , the calculation of the Z axis measurement is optional. In some embodiments, the calculation related to the Z axis sensor component may be omitted to simplify the calculation process. When the device **120** is positioned in the vehicle **110**, the position of the device **120** is not necessarily aligned with the horizon or the XY plane. While generally, forward direction X_V and lateral direction Y_V of the vehicle **110** are aligned or parallel to the horizon, the device **120**, in most cases, is going to be tilted with respect to the XY plane. This adds the complexity of calculating the misalignment angle θ_m between the vehicle **110** and the sensor frame of the device **120**. In one embodiment, to calculate the misalignment angle θ_m , the tilt direction which may also be calculated based on pitch, yaw, and roll of the device **120** is considered. In addition to the tilt of the sensor frame, the direction, angular velocity, angular acceleration, and any other suitable parameters of the device **120** are calculated using accelerometers and gyroscopes to extract the horizontal acceleration components. X_S is indicative of the horizontal component of the forward direction (or heading direction) of the sensor frame included in the device **120**. Y_S is indicative of the horizontal component of the lateral direction (or side direction) of the sensor frame included in the device **120** which is perpendicular to X_S . In one or more embodiments, X_S and Y_S are vectors of the sensor frame of the device **120**.

FIG. 1B is a view **105** of the vehicle **110** entering into a tunnel **130**, which typically has low or no GPS/GNSS signals being received by the device **120** or the user's mobile phone such that the rider may have inaccurate information about the location of the driver. It is noted that this disclosure may be applied to emergency vehicles and their headquarters or other applications where accurate vehicle location in low or no GPS/GNSS environments is desired. Entering into a building or reception of GPS/GNSS signals causes the device **120** to lose reception of GPS/GNSS signals or receive weak signals below a threshold that is capable of location detection and the location of the user inside the vehicle **110** may soon be lost.

By calculating the misalignment angle θ_m , the device's **120** inertial navigation system can accurately utilize the sensor data to compute accurately the location of the device **120** and the vehicle **110**, even if the GPS/GNSS signals are either lost or weak, by using the sensors included in the device. In this embodiment, the device **120** is aligned with the horizon or the XY plane. If the device **120** is leveled, it puts the X_V of the vehicle and X_S of the sensor frame of the device **120** in the same XY plane, thus making the calculation of the misalignment angle θ_m between X_V and X_S easier. However, in most cases, as explained, the device **120** is likely to have a tilt.

In one embodiment, to determine the degree of tilt of the device **120**, the positional relationship of the device **120** in terms of X, Y, Z components are determined using a gyroscope and an accelerometer of the device **120**. After the degree of tilt of the device **120** is determined, only the horizontal components of X, Y are extracted for the purpose of calculating the misalignment angle θ_m . However, as mentioned, in some embodiments the Z axis components can be considered in calculating the misalignment angle θ_m . In other embodiments, GPS sensor can be optionally used to receive the X_S , X_V component as well as the other X, Y, Z components. Yet in other embodiments, the X_V component of the vehicle may be received directly from the odometer of the vehicle **110** or any other movement related sensors of the vehicle **110**. By receiving speed, direction, angular velocity, angular acceleration information from the vehicle **110** through wireless or non-wireless means, this will reduce the calculation performed by the device **120**. The algorithms for calculating the misalignment angle θ_m based on data received from the IMU will be detailed in later figures.

The device **120** as used herein that is suitable for including various sensors. An example of these sensors may include accelerometers or gyroscopes or odometers or any other sensors for detecting motion related data of the device **120** and the vehicle **110**. Examples of motion related data may include, but are not limited to, direction, acceleration rate, angular velocity, angular acceleration, etc. The device **120** is can be a mobile device such a smart phone, tablet PC, GPS navigation device, or the like. In other embodiments, the device **120** can also be a separate standalone device manufactured for use in dead reckoning when GPS/GNSS signals are lost, such as the device shown in FIGS. **11A** and **11B**. Yet in other embodiments, the device **120** can be installed in a vehicle from the outset. In this case, the device **120** will be built into the vehicle and will be installed by the car maker in a known position. The device **120** may remain in the vehicle for the whole life of the vehicle. For these cases, factory calibration is applicable and this reduces many complex calculations related to the algorithm for computing the misalignment angle because the position of the device **120** within the vehicle **110** would be known.

The vehicle **110** as shown in FIGS. **1A** and **1B** indicates a car, but the application of the algorithm according to the present disclosure is not limited to vehicles and it may also be applicable to pickup truck or any other transportation vehicles. In other embodiments, the vehicle **110** may refer to drones, unmanned aerial vehicle (UAV), helicopters, boats, ships or air planes based on the application of the algorithm.

FIG. **2** is a system flow diagram **200** showing determining the misalignment angle θ_m using the device **120** according to an exemplary embodiment of the present disclosure. In one embodiment, a misalignment estimation block **205** determines the misalignment angle θ_m between the sensor frame and the vehicle frame using a Singular Value Decomposition (SVD) estimation method. Under this estimation method,

the angle is determined by applying orthogonal transformation on the estimation of direction of motion using standalone inertial sensors in the device **120**. In one or more embodiments, the vehicle dynamics under this estimation are modelled by non-holonomic constraints of the vehicle. These non-holonomic constraints relate to the fact that the vehicle dynamics are generally restricted to a forward direction. For example, the dynamics of vehicle motion may be restricted to a forward direction (longitudinal) because during navigation, the vehicle generally accelerates or decelerates along a straight line. When the vehicle moves along a straight line or road, the large component of the total acceleration will consist of longitudinal direction acceleration at acceleration and deceleration. The lateral direction component, which is perpendicular to the longitudinal direction, is likely to be minimal since most of the movement of the vehicle would be focused on moving forward. Considering these constraints may reduce or simplify the variables, formulas and algorithms involved in the misalignment angle calculation process.

In another embodiment, the misalignment estimation block **205** determines the misalignment angle between the sensor frame and the vehicle frame using an Orthogonal Transformation Estimation (OTE) method. Under this estimation method, the angle is determined by applying orthogonal transformation on the estimation of direction of motion using standalone inertial sensors in the device **120** as well as acceleration information derived from GPS/GNSS sensors. The vehicle dynamics under this estimation, too, are modelled by non-holonomic constraints of the vehicle.

Yet in another embodiment, the misalignment estimation block **205** determines the misalignment angle between the sensor frame and the vehicle frame using a GNSS Direction Cosine Matrix (GNSS DCM) method. Under this estimation method, the angle is determined by using inputs from the inertial sensors in the device **120** and the vehicle rotation matrix (DCM). The method further minimizes the error between the acceleration observed by the inertial sensor and the acceleration derived from GPS/GNSS sensors.

In further embodiments, the misalignment estimation block **205** determines the misalignment angle between the sensor frame and the vehicle frame using a fusion of the SVD, OTE, and GNSS DCM estimation methods. Under this estimation method, the output from the three estimation methods is weighted using a weighing technique to obtain an improved and accurate angle estimate with increased reliability. By utilizing the fusion method of all three estimation methods, a reliable angle result can be produced even under GPS/GNSS signal denied environments.

Referring to FIG. 2, the three axes gyroscope sensor provides angular rate sensing in three orthogonal directions at block **210**. At block **220**, the three axes accelerometer sensor provides specific forces sensing in the three orthogonal directions. At block **230**, the GPS/GNSS sensor provides position, velocity and precise time (PVT) data. Acceleration and speed can be derived by the received velocity components from the GPS/GNSS sensor. In other embodiments, where GPS/GNSS signals are weak or lost, the GPS/GNSS data input may be optional depending on the environment. At block **240**, the motion status information of the vehicle **110** is also inputted to the SVD block **250**, OTE block **260**, GNSS DCM block **270** and the fusion block **280**. The motion status information of the vehicle can be categorized, for example, as 'stop,' 'forward straight,' 'turning,' 'reverse,' 'reverse straight,' etc. This motion status **240** can be obtained by processing accelerometer data using techniques known to a person in the related art.

In one embodiment, the gyroscope data **210**, the accelerometer data **220**, and the motion status **240** is inputted into the SVD block **250**. The details of the processing in the SVD block **250** will be explained in FIG. 3. After the SVD angle estimation is produced based on the processing in the SVD block **250**, the angle is inputted to the fusion block **280**. In other embodiments, the SVD angle estimation produced based on the processing in the SVD block **250** may be output as the final result **290** without having to process through the fusion block **280**.

In one embodiment, the gyroscope data **210**, the accelerometer data **220**, the GPS/GNSS data **230**, and the motion status **240** is inputted into the OTE block **260**. The details of the processing in the OTE block **260** will be explained in FIG. 4. After the OTE angle estimation is produced based on the processing in the OTE block **260**, the angle is inputted to the fusion block **280**. In other embodiments, the OTE angle estimation produced based on the processing in the OTE block **260** may be output as the final result **290** without having to process through the fusion block **280**.

In one embodiment, the gyroscope data **210**, the accelerometer data **220**, the GPS/GNSS data **230**, and the motion status **240** are inputted into the GNSS DCM block **270**. The details of the processing in the GNSS DCM block **270** will be explained in FIG. 5. After the GNSS DCM angle estimation is produced based on the processing in the GNSS DCM block **270**, the angle is inputted to the fusion block **280**. In other embodiments, the GNSS DCM angle estimation produced based on the processing in the GNSS DCM block **270** may be output as the final result **290** without having to process through the fusion block **280**.

In one or more embodiments, the fusion block **280** will receive the SVD angle estimation, the OTE angle estimation, the GNSS DCM angle estimation from the SVD block **250**, the OTE block **260**, the GNSS DCM block **270**, respectively. Based on the combination of these angles after a weighing process based on the motion status of the vehicle and the signal strength of each sensors, the fusion block **280** outputs an optimal misalignment angle with increased reliability.

It will be readily understood by a person of ordinary skill in the art that the blocks, units, or modules within the misalignment estimation block **205** may be implemented as a software program or an algorithm that is capable of being executed through a processor in the device **120** or any other circuitry, programmed module, or programmable logic device (PLD) in the device **120**. The processor may include any suitable processing unit capable of performing the embedded software or algorithm of each block or module, among other various functions of the device. For example, it may include, but is not limited to, mobile application processors, microprocessors implemented with electronic circuitry, PLD, programmable controller, programmable logic controller, a general purpose processor with necessary algorithms programmed to perform the function, an integrated circuit (IC) or the like.

FIG. 3 is a flow diagram **300** showing determining a misalignment angle θ_m under the singular value decomposition (SVD) estimation method according to an exemplary embodiment of the present disclosure. The SVD estimation block **250** included in the processor of the device **120** receives the motion status of the vehicle **240**, the gyroscope data **210** and the accelerometer data **220**. The processor of the device **120** may also use the GPS/GNSS data to determine the motion state of the vehicle but may be optional.

When a vehicle is moving straight, and in case of a perfect alignment where the misalignment angle θ_m is 0 degrees, the

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signal measured by the accelerometer should be only concentrated in an axis pointing forward, the forward axis, with a lateral axis sensing pure noise. On the other hand, in case of imperfect alignment where the misalignment angle θ_m is not 0 degrees, the amount of the forward acceleration will be sensed by the lateral axis. Accordingly, a vehicle's forward acceleration provides sufficient information to estimate the angle offset between vehicle frame and sensor frame. A detailed explanation for calculating the misalignment angle θ_m will be explained further below.

At the SVD estimation block **250**, the accelerometer data **220** is received. The accelerometer data **220** includes the accelerometer data in the X direction (X Acc), the accelerometer data in the Y direction (Y Acc), and the accelerometer data in the Z direction (Z Acc). In one or more embodiments, the accelerometer in the device **120** is capable of receiving the X Acc, Y Acc, and Z Acc data of the vehicle **110**. For example, if the vehicle **110** is moving in a straight, forward direction, the acceleration components of the X direction (assuming that the vehicle is driving towards a hypothetical X direction) will be significant compared to the Y Acc, Z Acc components. If the vehicle **110** is making a turn (e.g., left, right or U-turn), the acceleration components of the +Y or -Y direction (assuming that the vehicle is turning towards the Y direction) will be significant compared to the X Acc, Z Acc components. If the vehicle **110** is going uphill or downhill, the acceleration components of the +Z or -Z direction (assuming that the vehicle is moving towards the Z direction) will be significant compared to the X Acc, Y Acc components.

At the SVD estimation block **250**, the gyroscope data **210** is also received. The gyroscope data **210** includes the data from the gyroscope in the X direction (X Gyro), the data from the gyroscope in the Y direction (Y Gyro), and the data from the gyroscope in the Z direction (Z Gyro). In one or more embodiments, the gyroscope in the device **120** is capable of receiving the X Gyro, Y Gyro, and Z Gyro data of the device **120**. In some embodiments, the sensor frame of the device **120** may be detected using the gyroscope signals X Gyro, Y Gyro, Z Gyro to calculate the tilt of the device **120** with reference to the vehicle **110**. Based on the gyroscope signals, the accelerometer signals X Acc, Y Acc, Z Acc can be tilt compensated at block **320**. These tilt compensated signals are then finally filtered in order to select the straight horizontal parts of the accelerometer signals X Acc, Y Acc at block **325**. In other embodiments, the GPS/GNSS sensors can be used separately or in conjunction with the accelerometer or other sensors if the GPS/GNSS signals are available.

Referring back to FIG. 3, at the straight line detection/significant motion detection block **310**, the processor of the device **120** determines the motion status **240** of the vehicle, based on the gyroscope data **210** and the accelerometer data **220**. In some embodiments, the processor of the device **120** may use the gyroscope data **210** to determine whether the vehicle **110** is moving in a straight line or moving significantly in a non-straight line. As with the other embodiments, the use of the GPS/GNSS data may be optional.

At the detection block **310**, whether the vehicle is moving in a straight line or in a significant motion (turn) is determined by the processor. In order to determine the movement of the vehicle, the processor receives sensor input from the gyroscope of the device **120**. If the vehicle **110** accelerates or decelerates while navigating along a straight line, the gyroscope will not measure significant angular velocity. Further, the value of the gyroscope may be indicative of the amount of rotation applied to the device **120** in the vehicle

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110. Based on the sensed data from the gyroscope and considering the angular velocity, the processor may determine that the vehicle **110** is in a straight line movement and the vehicle **110** is turning. If the vehicle **110** is travelling along a path that is not straight, for example, taking a left or right turn, or a U-turn, the gyroscope will be able to sense change in angle or angular velocity by the turn. In this case, the processor may determine that the vehicle **110** is in a significant motion (turn) rather than a straight line movement. If the gyroscope senses a significant angular velocity value below a threshold value, the processor of the device may conclude that there is no turn movement in any direction. The angular acceleration sensed by the gyroscope has to be higher than the threshold value (e.g., a threshold that indicates if the value is below this threshold value there may be no movement or insignificant movement, and if the value is above this threshold value there may be movement), for the processor to determine that the vehicle is not stationary (or also moving at a constant velocity) and accelerating/decelerating.

In other embodiments, the accelerometer data **220** may be used. The processor may use the gyroscope data **210** in combination with the accelerometer data **220**. If the device **120** senses an acceleration value below a threshold value indicating that there is acceleration movement in any direction, the processor may determine that the vehicle **110** is stationary or moving at a constant velocity. In reality, unless the vehicle **110** is at a complete stop, a noise value of the acceleration in any direction may be present. Accordingly, an acceleration value above the threshold value will be indicative of acceleration or deceleration. In some embodiments, the processor may determine, based on the accelerometer data **220** and the gyroscope data **210**, to determine the vehicle's straight line movement or significant movement.

In the tilt compensation block **320**, the accelerometer is used to determine the tilt direction of the device **120**. In one embodiment, the roll and pitch angle of the device **120** is calculated to determine the tilt direction. The roll and pitch angle of the device **120** or, more exactly, the sensor frame of the device **120**, is detected using the accelerometer data. Based on the accelerometer data **220**, the accelerometer signals X Acc, Y Acc, Z Acc are tilt compensated for the purpose of extracting the horizontal components of the accelerometer signals. In other embodiments, the gyroscope signals X Gyro, Y Gyro, Z Gyro is used in conjunction with the accelerometer signals X Acc, Y Acc, Z Acc for calculating the tilt direction and tilt compensation at block **320**.

In the horizontal acceleration block **325**, these tilt compensated signals are received and they are filtered in order to select the straight horizontal parts of the signals. Based on the accelerometer signals X Acc, Y Acc, Z Acc, the horizontal acceleration component HX Acc, HY Acc, HZ Acc are extracted after considering the tilt of the device **120** (or more accurately the tilt of the sensor frame). In some embodiments, the gyroscope signals X Gyro, Y Gyro, Z Gyro may be used for tilt compensation. These gyro signals are inputted to calculate the tilt and the calculated tilt direction is used to extract the straight horizontal parts of the accelerometer signals X Acc, Y Acc, Z Acc at block **325**.

In the data filtering block **315**, the motion status information processed from the motion detection block **310** and the input from the gyroscope data **210** are combined with the accelerometer data **220** and the information processed through the horizontal acceleration block **325**. At the data filtering block **315**, any noise or data irrelevant to the analysis of the misalignment angle is filtered out. In some

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embodiments, the filter may have a certain filter range for passing signals that fall within that range and filtering out the rest of the signals. For example, any random noise signals not indicative of accurately reflecting the vehicle motion may not be considered. This filtered out data will not be passed along to the buffer **330** for storing as it may be likely to contain noise. In other embodiments, the data filtering block **315** may remove acceleration when there is no sufficient movement. That is, if the acceleration value is close to zero (which means there is not any significant movement), it will remove the acceleration value. The data filtering block **315** extracts only the accelerometer data when significant acceleration is detected to ensure that irrelevant data or noise data is not input to the buffer block **330**.

After data filtering is performed at the data filtering block **315**, the block **315** outputs the filtered data to the buffer **330**. In the buffer **330**, if the data is enough for analyzing the misalignment angle, the data in the buffer **330** will be passed to the auto-correlation matrix block **345** for computation. The processor may determine whether the data in the buffer **330** exceeds the minimum threshold amount for calculating the misalignment angle to produce a reliable result. For example, the data in the buffer **330** may be accumulated to the point that it will produce a misalignment angle between the IMU axes and vehicle longitudinal direction with an accuracy better than 2 degrees. In other embodiments, the buffer may be updated when the accuracy is to be better than 1 degree.

On the other hand, if the amount of data accumulated in the buffer **330** is not enough, the process proceeds to block **335** to update the buffer. The buffer will be updated by receiving renewed sensor data for the next epoch **340** or based on a selected interval of time. The buffer after being updated based on the data received in the next epoch may provide the information regarding vehicle motion (e.g., acceleration, tilt direction, etc.) from the IMU to the buffer **330**. The processor may determine whether the buffer **330** has enough data for calculating the misalignment angle to produce a reliable result. If the processor determines that the buffer **330** has not met the threshold indicating sufficient minimum data for misalignment calculation, the processor will not proceed with the misalignment angle estimation based on the accumulated data and will update the buffer again at block **335**.

At the autocorrelation matrix block **345**, the processor computes the input data received using an autocorrelation matrix. In one embodiment, the two accelerations in the horizontal plane, X Acc, Y Acc, are input to the autocorrelation matrix block **345**. During or after computing using the autocorrelation matrix, a SVD method is used at the SVD block **350** to minimize the correlation between the two X, Y components of accelerometer signal (e.g., acceleration vector). Here, the input to the SVD block **350** is a cross correlation matrix (2x2) between X axis acceleration and the Y axis acceleration. That is, the autocorrelation matrix block **345** and the SVD block **350** proceed based on the below process.

$$\begin{bmatrix} X \text{ Acc} \\ Y \text{ Acc} \end{bmatrix}_{2 \times N} \rightarrow \text{Auto Correlation Block} \rightarrow$$

$$\begin{bmatrix} \sum X \text{ Acc} * X \text{ Acc} & \sum X \text{ Acc} * Y \text{ Acc} \\ \sum Y \text{ Acc} * X \text{ Acc} & \sum Y \text{ Acc} * Y \text{ Acc} \end{bmatrix}_{2 \times 2} \rightarrow \text{SVD}$$

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After angle estimation is produced based on the SVD method, the misalignment angle is checked as to the reliability of the results. At angle estimation and quality check block **355**, the reliability of the angle is tested to minimize the error and increase the level of quality of the misalignment angle. In one embodiment, at quality check block **355**, the processor computes the quality of the output and filters any poor estimates. By filtering out the poor estimates, it improves the quality and the reliance of the final output. When the processor determines that the quality is poor at quality check block **355**, it can receive input from the next epoch **340** and update the buffer **335** again to receive sufficient data for computation. As previously mentioned, the determination on whether the quality is good or poor may depend on the produced misalignment angle between IMU axes and vehicle longitudinal direction having an accuracy better than 2 degrees. This means that the calculation solely based on IMU will not deviate more than 2 degrees from the actual vehicle route in a GPS/GNSS signal absent environment.

In one or more embodiments, the process in quality check block **355** includes checking the variance ratio of the rotated X axis (the forward axis) and the Y axis (the lateral axis) using SVD. For example, since the acceleration in the forward axis is high, the variance should be high compared to the lateral axis. If the ratio of variance in the forward axis and the lateral axis is high, this indicates that the quality of the SVD is good. In other embodiments, the process in quality check block **355** includes checking the minimum variance in the forward axis. Yet in other embodiments, the process in quality check block **355** includes checking the minimum number of samples N (e.g., N in 2xN data).

In some embodiments, a quality check is performed to ensure the accuracy and the quality of the produced misalignment angle. This check is performed at quality check block **355**. In other embodiments, the quality check may be performed during any suitable stage in order to maintain the integrity and the accuracy of the data. The data check process is not limited to being performed at only certain stages of the algorithm. Once quality is confirmed, the SVD angle is output at block **360**. The misalignment angle based on the SVD method, along with the X and Y component of the horizontal acceleration, may be stored in cumulative form to minimize the storage. For example, the data may be stored in the form of a mean, variance, cross-correlation, or the like.

FIG. 4 is a flow diagram **400** of determining a misalignment angle θ_m under the orthogonal transformation estimation (OTE) method according to an exemplary embodiment of the present disclosure. In one embodiment, the OTE block **260** included in the processor of the device **120** receives the motion status of the vehicle **240**, the gyroscope data **210** and the accelerometer data **220**. As with other embodiments, the processor of the device **120** may use the GPS/GNSS data **230** as an option. For the sake of simplicity, blocks that perform similar functions compared to the SVD estimation block **250** will be omitted. Blocks or modules named the same or similar as FIG. 3 will generally function in the same or similar manner. The same or similar functionality will not be explained here in order not to be repetitive.

At the straight line detection/significant motion detection block **410**, the processor of the device **120** determines the motion status **240** of the vehicle, based on the gyroscope data **210**. In one embodiment, the gyroscope data **210** along with the motion status of the vehicle **240** is used to detect straight road sections.

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In the tilt compensation block **420**, the accelerometer is used to determine the tilt direction of the device **120**. Once the tilt compensation is calculated in block **420**, the results of the accelerometer signals, X Acc, Y Acc, and Z Acc are output to the horizontal acceleration block **425**.

In the horizontal acceleration block **425**, the horizontal acceleration component HX Acc, HY Acc, HZ Acc are extracted based on the tilt compensated signals. In some embodiments, the gyroscope signals X Gyro, Y Gyro, Z Gyro may be used for tilt compensation. These gyro signals are inputted to calculate the tilt and the calculated tilt direction is used to extract the straight horizontal parts of the accelerometer signals.

In the data filtering block **415**, the motion status information processed from the motion detection block **410** and the input from the gyroscope data **210** are combined with the accelerometer data **220** and the information processed through the horizontal acceleration block **425**. At the data filtering block **415**, the GPS/GNSS data **230** is also considered. In one or more embodiments, the GPS/GNSS data **230** may be used in conjunction with the motion detection block **410**, or block **420**, or block **425** to improve the calculation when the GPS/GNSS signals are present.

After data filtering is performed at the data filtering block **415**, the data filtering block **415** outputs the filtered data to the buffer **430**. In the buffer **430**, if the data is enough for analyzing the misalignment angle, the data in the buffer **430** will be passed to the cross-correlation matrix block **445** for computation. The processor may determine whether the data in the buffer **430** exceeds the minimum threshold amount for calculating the misalignment angle to produce a reliable result. For example, the data in the buffer **430** may be accumulated to the point that will produce a misalignment angle between IMU axes and vehicle longitudinal direction with accuracy better than 2 degrees. That is, a misalignment angle between sensor frame (X_s) and the vehicle frame (X_v) with accuracy better than 2 degrees. In other embodiments, the buffer may be updated when the accuracy is to be better than 1 degree.

On the other hand, if the amount of data accumulated in the buffer **430** is not enough, the step proceeds to block **435** to update the buffer. The buffer will be updated by receiving renewed sensor data for the next epoch **440**. The processor may determine whether the buffer **430** has enough data for calculating the misalignment angle to produce a reliable result. If the processor determines that the buffer **430** has not met the threshold indicating sufficient minimum data for misalignment calculation, the processor will not proceed with the misalignment angle estimation based on the accumulated data and will update the buffer again at block **435**.

At the cross-correlation matrix block **445**, the processor computes the data received using a cross-correlation matrix. The cross-correlation between a cross-correlation transformation matrix and the input from the GPS/GNSS data **230** is computed. In block **450**, the orthogonal transformation estimation is applied between the horizontal X Acc, Y Acc and the GPS/GNSS data **230**. At the orthogonal transformation estimation block **450**, the orthogonal transformation minimizing the correlation between horizontal X, Y acceleration components (e.g., X_s and Y_s of the sensor frame of the device **120**, X_v and Y_v of the vehicle frame of the vehicle **110**) and GPS/GNSS is found. When the orthogonal transformation minimizing the correlation between the horizontal X, Y acceleration components and GPS/GNSS is found, an orthogonal transformation matrix is applied to produce the misalignment angle.

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In other embodiments, the two X, Y components of accelerometer signal after being processed using the OTE method, may refer to the horizontal value of the vehicle frame (X_v and Y_v) and the sensor frame (X_s and Y_s).

In one or more embodiments, by using the orthogonal transformation matrix, the relationship between the acceleration component of the vehicle frame and the acceleration component of the sensor frame may be established as follows:

$$Acc_{VEHICLE} = R * Acc_{SENSOR} \quad (R \text{ denotes the orthogonal transformation matrix})$$

The degree of rotation of the orthogonal transformation represents the misalignment angle θ_m .

The horizontal components of the vehicle X_v and Y_v may be represented using the below matrix

$$Acc_{VEHICLE} = \begin{bmatrix} X_{Acc_Vehicle} \\ Y_{Acc_Vehicle} \end{bmatrix}$$

In addition, the horizontal components of the sensor frame X_s and Y_s may be represented using the below matrix

$$Acc_{SENSOR} = \begin{bmatrix} X_{Acc_Sensor} \\ Y_{Acc_Sensor} \end{bmatrix}$$

By applying the orthogonal transformation matrix according to the aforementioned $Acc_{VEHICLE} = R * Acc_{SENSOR}$ formula,

$$R = \begin{bmatrix} \cos(\text{Angle}) & \sin(\text{Angle}) \\ -\sin(\text{Angle}) & \cos(\text{Angle}) \end{bmatrix}, \text{ where Angle} = \text{misalignment angle } \theta_m.$$

The following equation can be established:

$$\begin{bmatrix} X_{Acc_Vehicle} \\ Y_{Acc_Vehicle} \end{bmatrix} = \begin{bmatrix} \cos(\text{Angle}) & \sin(\text{Angle}) \\ -\sin(\text{Angle}) & \cos(\text{Angle}) \end{bmatrix} * \begin{bmatrix} X_{Acc_Sensor} \\ Y_{Acc_Sensor} \end{bmatrix}$$

The calculation of the above equation produces the misalignment angle θ_m between the vehicle frame and the sensor frame.

After the angle estimation is produced based on the OTE method, the misalignment angle is checked as to the reliability of the results. At the angle estimation and quality check block **455**, the reliability of the angle is tested to reduce the error and increase the level of quality of the misalignment angle. In one embodiment, at block **455**, the processor computes the quality in the output and filters poor estimates. By filtering out the poor estimates, it improves the quality and the reliance of the final output. When the processor determines that the quality is poor at block **455**, it can receive input from the next epoch **440** and update the buffer **435** again to receive sufficient data for computation. As previously mentioned, the determination on whether the quality is good or poor may depend on aforementioned accuracy angle levels (e.g., 2 degrees).

Once quality is confirmed, the OTE angle is output at block **460**. The misalignment angle based on the OTE method, along with the X and Y component of the horizontal acceleration, may be stored in cumulative form to minimize the storage as with other embodiments.

FIG. 5 is a flow diagram 500 of determining a misalignment angle θ_m under the GNSS direction cosine matrix (DCM) method according to an exemplary embodiment of the present disclosure. In one embodiment, the GNSS DCM estimation block 270 included in the processor of the device 120 receives the motion status of the vehicle 240, the gyroscope data 210, the accelerometer data 220, and the GPS/GNSS data 230.

At the sensor calibration model block 510, the accelerometer data 220 is received. Here, accelerometer measurement is tilt (e.g., pitch and roll) compensated to yield an IMU horizontal acceleration. In one embodiment, the sensor calibration model block 510 may calculate the horizontal acceleration component HX Acc, HY Acc by using the accelerometer data 220. In other embodiments, the sensor calibration model block 510 may calculate the horizontal acceleration components by tilt compensation using gyroscope signals X Gyro, Y Gyro, Z Gyro from the gyroscope data 210. The sensor calibration model block 510 may extract horizontal acceleration components based on the tilt compensated signals. These gyro signals may be inputted to sensor calibration model block 510 to calculate the tilt and the calculated tilt direction is used to extract the straight horizontal parts of the accelerometer signals X Acc, Y Acc.

Based on the above calculation method, the sensor calibration model block 510 outputs the calibrated acceleration data for the X acceleration component and the Y acceleration component. This calibrated acceleration data in the X axis and the Y axis is actually sensed data compared to the predicted acceleration data in the X axis and the Y axis output from the vehicle to sensor frame DCM rotation block 530 that is produced based on the calculation from the angle estimation minimizing acceleration error block 520.

At the angle estimation minimizing acceleration error block 520, the processor of the device 120 receives the motion status 240 of the vehicle and the gyroscope data 210. Based on these inputs, the angle estimation minimizing acceleration error block 520 outputs an angle in the time domain at time t, namely angle (t). In one embodiment, the angle estimation minimizing acceleration error block 520 is implemented using a Kalman Filter. In block 520, the Kalman filter predicts the angle in order to minimize or substantially reduce the error between the acceleration measured by the IMU sensors in the sensor frame and the GNSS acceleration rotated to the sensor frame. The Kalman Filter estimates an angle at which both the acceleration measured from the IMU sensors and the GPS/GNSS sensors are similar. The angle that produces the acceleration measured from the IMU sensors and the GPS/GNSS sensors that are similar indicates that this angle has the least or minimum error. After the process at block 520, the angle in the time domain, angle (t), is then inputted into the inverse Z transform block 540. The inverse Z transform, Z^{-1} , outputs an angle (t-1), an angle at time t-1.

In the vehicle to sensor frame DCM rotation block 530, the GPS/GNSS data 230 is received. The GPS/GNSS data 230 includes the speed of the vehicle determined using the GPS/GNSS sensor, GNSS speed. The GPS/GNSS data 230 also includes the forward acceleration and other acceleration information. In one embodiment, at the vehicle to sensor frame DCM rotation block 530 the processor applies a rotation matrix to the inputs. For example, the block 530 rotates the forward and lateral acceleration measured by the GPS/GNSS sensors to X and Y acceleration in the sensor frame produced by angle estimation. In one embodiment, the rotation is represented as follow:

$$\begin{bmatrix} X \text{ Acc_Sensor} \\ Y \text{ Acc_Sensor} \end{bmatrix} = \begin{bmatrix} \cos(\text{Angle}) & -\sin(\text{Angle}) \\ \sin(\text{Angle}) & \cos(\text{Angle}) \end{bmatrix} * \begin{bmatrix} X \text{ Acc_Vehicle} \\ Y \text{ Acc_Vehicle} \end{bmatrix}.$$

After the calculation at the vehicle to sensor frame DCM rotation block 530, the processor outputs a predicted X acceleration component and the Y acceleration component based on angle (t-1) outputted from the Z^{-1} transform and the GPS/GNSS data 230. In one embodiment, the predicted or considered accelerations may match the same reference frame of the sensor frame of the IMU when ported from the vehicle frame. Accelerations from the GPS/GNSS navigation sensors are ported to the sensor frame of the device 120 using a rotation matrix using the misalignment angle and tilt direction of the device 120. Any difference between the sensor frame and the vehicle frame is proportional to the misalignment angle. As explained, in the vehicle to sensor frame DCM rotation block 530, accelerations detected based on the GPS/GNSS sensor is rotated by the misalignment angle to port them into the sensor frame data outputted from the sensors calibration model block 510, and the results of the actually sensed X, Y acceleration component and the predicted X, Y acceleration component are compared.

In the angle estimation minimizing acceleration error block 520, the processor receives the acceleration error based on the difference between the actually sensed X, Y acceleration component and the predicted X, Y acceleration component. The processor estimates the optimum angle, reducing the root mean square error between the GPS/GNSS acceleration (prediction) and the IMU acceleration (actual measurement) at the angle estimation minimizing acceleration error block 520.

The GNSS DCM estimation block 270 calculates based on comparing the horizontal acceleration measured independently by two sources, the IMU sensor in the device 120 and the GPS/GNSS sensor which may be located in the vehicle 110. In some embodiments, the GPS/GNSS sensor may be located within the device 120. After an error between the predicted acceleration value and the sensed acceleration value is minimized using various filters including the Kalman Filter (KF), Infinite Impulse Response (IIR) filter, and LS filters, the misalignment angle with a reduced error is output at block 550. The filter that may be used for reducing or minimizing the angle estimation error is not limited to the KF, IIR, LS filters, and a person of ordinary skill in the art would readily understand that other suitable filters can be applied to obtain substantially the same or similar results. As explained, the implementation of block 520 was performed by using a Kalman Filter, but other filters may be applied to achieve the same or similar results.

FIG. 6 is a flow diagram 600 of a fusion block 280 of the outputted misalignment angles according to an exemplary embodiment of the present disclosure. The processor of the device 120 receives the SVD angle calculated from the SVD estimation block 250, the OTE angle calculated from the OTE block 260, the GNSS DCM angle calculated from the GNSS DCM estimation block 270. The output from the three estimation methods is merged to get a unique and improved angle estimate with increased reliability. In one or more embodiments, the fusion is operated by a Kalman Filter and can work in all possible scenarios, including a GPS/GNSS signal denied environment.

At block 610, the acceptable SVD angle conditions are considered before applying the angle weight at block 620. The acceptance SVD angle conditions block 610 determines whether the received SVD angle is within an acceptable

range that will produce a reliable misalignment angle. Acceptance conditions specific for each input are applied at block 610. At calculation of the SVD angle weight block 620, the processor determines how much weight will be applied based on multiple factors. For example, the factors may include, but are not limited to, the presence or absence of certain sensors, the signal strength received based on the sensors, the motion characteristics of the vehicle, the path conditions the vehicle is being driven on, etc. In other embodiments, the calculation SVD angle weight block 620 outputs confidence and weight which reflect how accurate or erroneous the output is. In blocks which may be implemented using Kalman Filters, the measurement of the SVD angle made by applying the Kalman Filter is associated with uncertainty that is proportional to the weights. For example, if the uncertainty is great, the value of the produced angle (the SVD angle) may be given less weight. On the other hand, if the uncertainty is less, the value of the produced SVD angle may be given greater weight than the OTE angle or the GNSS DCM angle. Once it determines the weight, the SVD angle is given a weight factor, W_{SVD} .

At block 630, the acceptable OTE angle conditions are considered before applying the angle weight at block 640. The acceptance OTE angle conditions block 630 determines whether the received OTE angle is within an acceptable range that will produce a reliable misalignment angle. Acceptance conditions specific for each input are applied at block 630. At calculation of the OTE angle weight block 640, the processor determines how much weight will be applied based on multiple factors. In other embodiments, the calculation OTE angle weight block 640 outputs confidence and weight which reflect how accurate or erroneous the output is. The blocks which may be implemented using Kalman Filters, the measurement of the OTE angle by applying the Kalman Filter is associated with uncertainty that is proportional to the weights. For example, if the uncertainty is great, the value of the produced angle (the OTE angle) may be given less weight. On the other hand, if the uncertainty is less, the value of the produced OTE angle may be given greater weight. Once it determines the weight, the OTE angle is given a weight factor, W_{OTE} . As explained in connection with block 620, the Kalman Filter can be implemented based on the above mentioned formula and relationship between variables.

At block 650, the acceptable GNSS DCM angle conditions are considered before applying the angle weight at block 660. The acceptance GNSS DCM angle conditions block 650 determines whether the received GNSS DCM angle is within an acceptable range that will produce a reliable misalignment angle. Acceptance conditions specific for each input are applied at block 650. At calculation of the GNSS DCM angle weight block 660, the processor determines how much weight will be applied based on multiple factors. In other embodiments, the calculation of the GNSS DCM angle weight block 660 outputs confidence and weight which reflect how accurate or erroneous the output is. As explained, the blocks may be implemented using Kalman Filters, and the measurement of the GNSS DCM angle by applying the Kalman Filter is associated with uncertainty that is proportional to the weights. For example, if the uncertainty is great, the value of the produced angle (the GNSS DCM angle) may be given less weight. On the other hand, if the uncertainty is less, the value of the produced GNSS DCM angle may be given greater weight. Once it determines the weight, the GNSS DCM angle is given a weight factor, W_{GNSS_DCM} . As explained in connection with

blocks 620 and 640, the Kalman Filter can be implemented based on the above mentioned formula and the relationship between variables.

At the weight based merging block 670, the processor receives the motion status 240 and the SVD angle, OTE angle, GNSS DCM angle, and respective weight W_{SVD} , W_{OTE} , W_{GNSS_DCM} . The fusion method applied at the weight based merging block 670 combines the output of the misalignment angle. The combination can rely on different techniques such as the use of KF filters, adaptive IIR filters. However, the fusion method is not limited to these filters. In one embodiment, the KF approaches can be used to estimate the final misalignment angle by minimizing the root mean square error between the measurement data and the predicted data. In other embodiments, the fusion method provides a final permanent estimate of the misalignment angle based on a set of selected metrics minimization. However, a person of ordinary skill in the art would appreciate that other methods may be used to achieve similar results.

After the misalignment angles are weighted, the merged angle estimation is inputted to the finalize estimation block 680. In one embodiment, at the finalize estimation block 680, the algorithm keeps monitoring confidence to accept or reject the new merged misalignment angle estimation. Since the misalignment angle may remain constant for a short duration, the processor may reject any poor misalignment angle estimates. In some embodiments, the processor may refine the current estimation of the misalignment angle by taking renewed angles and renewed weight based on the confidence. In other embodiments, at the finalize estimation block 680, the confidence and stability are monitored over time to produce a reliable angle final estimation.

FIG. 7 is a flow diagram 700 of determining a misalignment angle θ_m under the SVD estimation method according to another embodiment of the present disclosure. In one embodiment, the SVD estimation block 250 included in the processor of the device 120 receives the gyroscope data 210 and the accelerometer data 220. In some embodiments, the processor of the device 120 may also use the GPS/GNSS data 230 to determine the motion state of the vehicle 110. Three motion states of the vehicle, straight line detection 710, static detection 715, and significant motion detection 720, will be explained.

The processor of the device 120 may detect at static detection block 715 whether the vehicle is stationary or moving. At static detection block 715, in order to determine whether a vehicle is stationary or moving, the processor receives sensor input from the accelerometer and the gyroscope of the device 120. Static detection is important to initialize gravity vector and gyro bias computation. If the accelerometer in the device 120 senses an acceleration value below a threshold value indicative of an acceleration movement in any direction, the processor may determine that the vehicle 110 is stationary or moving at a constant velocity. In reality, unless the vehicle is at a complete stop, a noise value of the acceleration in any direction may be present. In some embodiments, the processor may determine whether the vehicle 110 is stationary or moving based on the comparison of the value of the accelerometer data 220 received and the threshold value. If the accelerometer senses angular acceleration above a threshold value, the processor may determine that the vehicle 110 is not moving in a straight line. Further, to confirm the motion state of the vehicle 110, the processor may use the gyroscope data 210 in combination with the accelerometer data 220. In some embodiments, the

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processor may consider both the gyroscope data **210** and the accelerometer data **220** at the same time, or analyze the data one at a time in any order.

The straight line detection is performed primarily to extract segments of measurement data where the vehicle is moving in a straight line and no significant lateral acceleration is observed. The straight line detection is driven by tests on negligible (above threshold) yaw rate condition over fixed window length. At the straight line detection block **710**, the processor may confirm that the vehicle **110** is moving in a straight line by further utilizing the gyroscope data **210** along with the accelerometer data **220**. For example, the gyroscope data **210** may show information about the motion status of the vehicle **110** when the vehicle accelerates or decelerates. When the vehicle **110** accelerates, the gyroscope of the device **120** can pick up the acceleration information because the device **120** will likely tilt together even though it may fixated in a certain location in the vehicle **110** due to the acceleration. Gyro bias may also be considered in connection with analyzing the motion status of the vehicle **110**. In some embodiments, after the initial determination of the motion status of the vehicle at the static detection block **715**, the processor may proceed to the straight line detection block **710** for further determination of the vehicle movement.

At the significant motion detection block **720**, the processor determines that the motion dynamics are significant based on the accelerometer data **220** and the GPS/GNSS data **230**. Significant motion detection block **720** is used to extract sensor data during motion when acceleration (above a threshold) is present in the longitudinal direction. When the vehicle is moving at constant speed motion, there is not sufficient information available to help with determination of the misalignment angle. During acceleration or deceleration, sensor data for this motion contains sufficient information about motion direction. In one embodiment, the significant motion is detected using the GPS speed, if GPS available. In one embodiment, the module checks the change in speed from the last 2 seconds or total horizontal acceleration in 1 second. Here, if the vehicle **110** is stationary, the gravity vector may be updated with high weight and the gyro bias may be computed. If the vehicle **110** is not moving in a straight line or no significant motion detected, the processor may wait for the next measurement epoch.

Referring back to the significant motion detection block **720**, the processor uses the accelerometer data **220** to determine if motion dynamics are significant to use it for calculation of the misalignment angle and confidence estimation. In other embodiments, the processor may use the GPS/GNSS data **230**, if available, to determine the motion dynamics and use it for calculating the misalignment angle and confidence estimation. In one embodiment, a significant motion dynamic may refer to an event of a vehicle **110** moving in a non-static manner while changing the direction of the vehicle **110**. For example, if the vehicle **110** is constantly making turns left to right, these motions may be referred to as being significant and will be sensed through the accelerometer. In other examples, a vehicle's motion may be significant when the angular velocity of the vehicle **110** changes frequently in a certain period of time. The value of the angular acceleration of the vehicle **110** may also be used in determining significant vehicle motion.

In the roll and pitch estimation block **705**, the accelerometer is used to determine the roll and pitch angle of the device **120**. The roll and pitch angle of the device **120** or, more exactly, the sensor frame of the device **120**, is detected using the gyroscope signals X Gyro, Y Gyro. Based on the

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gyroscope signals, the accelerometer signals X Acc, Y Acc are tilt compensated. In some embodiments, the Z axis component can be considered in tilt compensation. These tilt compensated signals are then finally filtered in order to select the straight horizontal parts.

At the determination of useful data block **725**, the processor aggregates the motion status information from the straight line detection block **710**, static detection block **715**, and the significant motion detection block **720**. The processor will determine the dominant motion status of the vehicle **110** at this stage and determine which analyzed data would be used for calculation. In one embodiment, the processor at the determination of useful data block **725** will calculate the noise amount of the data detected and if it falls shorter than the minimum threshold value, it may discard the data based on the determination that a significant amount of noise is included in the data. In another embodiment, the processor at the determination of useful data block **725** will give a weight based on the signal/noise ratio of each piece of data, and filter out irrelevant data that will not be used for calculating the misalignment angle. Further, at block **725**, predominant movement may be extracted for calculating the misalignment angle, and any bumps due to the road condition will also be filtered out as noise (e.g., even though a vehicle is moving in a straight line, bumps or ramps may cause the Z axis of the accelerometer to sense value; however, these noise values may be filtered out based on the dominant motion of the vehicle during a certain period of time or epoch). When the data is processed at block **725**, the data is inputted into the horizontal acceleration block **730** and the GPS derived forward and lateral acceleration block **735**.

In one embodiment, at the horizontal acceleration block **730**, the processor receives input from the roll and pitch estimation block **705**, the determination of useful data block **725** and the GPS/GNSS data **230**. In another embodiment, at the horizontal acceleration block **730**, the processor receives all of the input from the gyroscope data **210**, the accelerometer data **220**, the GPS/GNSS data **230**, as well as the input processed through the straight line detection block **710**, the static detection block **715**, the significant motion detection block **720**, the roll and pitch estimation block **705**, and the determination of useful data block **725**. In some embodiments, the processor may have only some of the sensed data at the horizontal acceleration block **730**. In some examples, if there is only a straight line movement in the vehicle, the data processed through the significant motion detection block **720** may be omitted or not used. In this block **730**, the processor extracts the horizontal acceleration component based on the data received. In some embodiments, the horizontal acceleration block **730** keeps this data in the buffer of the device **120**. If the amount of data reaches a threshold indicating sufficient minimum data for misalignment calculation, the processor will proceed with the misalignment angle estimation based on the accumulated data.

At the GPS derived forward and lateral acceleration block **735**, the processor calculates the forward and lateral acceleration of the vehicle **110** based on the input from the GPS/GNSS data **230** and the determination of useful data block **725**.

At the orthogonal transformation estimation block **740**, the processor may compute the inputs based on the data received from the horizontal acceleration block **730**. In one embodiment, an autocorrelation transformation matrix may be used. After computing using the autocorrelation transformation matrix, a SVD method is used to minimize the correlation between the two X, Y components of acceler-

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ometer signal (e.g., acceleration vector). An exemplary process is explained in a block below:

$$\begin{bmatrix} X_{Acc} \\ Y_{Acc} \end{bmatrix}_{2 \times N} \rightarrow \text{Auto Correlation Block} \rightarrow \begin{bmatrix} \sum X_{Acc} * X_{Acc} & \sum X_{Acc} * Y_{Acc} \\ \sum Y_{Acc} * X_{Acc} & \sum Y_{Acc} * Y_{Acc} \end{bmatrix}_{2 \times 2} \rightarrow SVD.$$

In one embodiment, the two X, Y components of accelerometer signal after being processed using the SVD method, may refer to the horizontal value of the vehicle frame (X_V and Y_V) and the sensor frame (X_S and Y_S).

In one or more embodiments, by using the above computation method, the relationship between the acceleration component of the vehicle and the acceleration component of the sensor frame may be established as follows:

$$Acc_{VEHICLE} = R * Acc_{SENSOR} \text{ (R denotes the rotation matrix).}$$

The rotation of the orthogonal transformation represents the misalignment angle θ_m .

The horizontal components of the vehicle X_V and Y_V extracted from the horizontal acceleration block **730** may be represented using the below matrix:

$$Acc_{VEHICLE} = \begin{bmatrix} X_{Acc_Vehicle} \\ Y_{Acc_Vehicle} \end{bmatrix}.$$

In addition, the horizontal components of the sensor frame X_S and Y_S extracted from the horizontal acceleration block **730** may be represented using the below matrix:

$$Acc_{SENSOR} = \begin{bmatrix} X_{Acc_Sensor} \\ Y_{Acc_Sensor} \end{bmatrix}.$$

By applying the rotation matrix according to the aforementioned $Acc_{VEHICLE} = R * Acc_{SENSOR}$ formula,

$$R = \begin{bmatrix} \cos(\text{Angle}) & \sin(\text{Angle}) \\ -\sin(\text{Angle}) & \cos(\text{Angle}) \end{bmatrix}, \text{ where Angle = misalignment angle } \theta_m.$$

The following equation can be established:

$$\begin{bmatrix} X_{Acc_Vehicle} \\ Y_{Acc_Vehicle} \end{bmatrix} = \begin{bmatrix} \cos(\text{Angle}) & \sin(\text{Angle}) \\ -\sin(\text{Angle}) & \cos(\text{Angle}) \end{bmatrix} * \begin{bmatrix} X_{Acc_Sensor} \\ Y_{Acc_Sensor} \end{bmatrix}.$$

The calculation of the above equation produces the misalignment angle θ_m between the vehicle and the sensor frame.

In some embodiments, by applying the orthogonal transformation to identify the axis orientation, it is possible to minimize the correlation between two acceleration vectors which are in the sensor frame. In other embodiments, the orthogonal transformation is applied such that the correlation of $Acc_{VEHICLE}$ can be minimized. After the orthogonal transformation estimation block **740** produces the misalignment angle, the misalignment angle is outputted to the Kalman Filter block **760**.

At the buffer block **745**, the GPS and accelerometer information is stored in a memory of the device **120**. At the

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computation block **755**, the rotation matrix that minimizes the root mean square deviation (RMSD) between the GPS data and accelerometer data is determined. This output is provided to the Kalman Filter (KF) block **760**. In one embodiment, valid measurements are inputted and fused in the KF block **760** to estimate optimum results. In one embodiment, the determination of useful data block **725** filters out any measurements that have significant noise components provided to the subsequent blocks valid measurements. For example, valid measurements may refer to measurements that are noise filtered and integrity of the data is above a threshold that ensures output misalignment angle having less than 2 degrees of error.

At the KF block **760**, inputs from the various blocks are analyzed and a misalignment angle output is produced. The details of the KF block **760** will be explained in detail later on.

At the diversity check block **765**, the processor may determine whether the produced misalignment angle output is consistent enough to move forward with the calculation of the misalignment angle. If the processor does not determine that the data is sufficient, the process will receive further data accumulation in the next epoch. In one or more embodiments, diversity is also checked using the Eigenvalue ratio.

At the misalignment angle and confidence estimation block **770**, the processor computes the confidence in the output which allows to filter poor estimates. By filtering out the poor estimates, it improves the quality and the reliance of the final output. In some embodiments, to minimize the error any suitable error minimization algorithm may be applied and without limitation to certain algorithms. In other embodiments, updated data may be received from the gyroscope data **210** or the accelerometer data **220**, and follow through the previously mentioned blocks to reduce the error in the misalignment angle. After a reliable misalignment angle is produced, the aforementioned algorithm proceeds to block **780** and exits the process.

FIG. **8** is a flow diagram **800** of an exemplary Kalman Filter according to one embodiment of the present disclosure. Kalman filtering is an algorithm that uses a series of measurements observed over time, containing statistical noise and other inaccuracies. The filter produces estimates of unknown variables that tend to be more accurate than those based on a single measurement alone, by estimating a joint probability distribution over the variables for each time-frame. In one embodiment, the algorithm works in a two-step process. In the prediction step, the Kalman filter produces estimates of the current state variables, along with their uncertainties (e.g., noise, inaccuracies, etc.). Once the outcome of the next measurement (necessarily corrupted with some amount of error, including random noise) is observed, these estimates are updated using a weighted average, with more weight being given to estimates with higher certainty. The algorithm may be recursive. In one example, it can run in real time, using only the present input measurements and the previously calculated state and its uncertainty matrix, and no additional past information is required. FIG. **8** shows one way of using the Kalman Filter according to the present disclosure. In one or more embodiments, a similarly designed Kalman Filter may be used in multiple blocks or modules for misalignment calculation. In addition, the Kalman Filter as depicted in FIG. **8** may be used to implement for the Kalman Filter block **760** in FIG. **7**.

In one embodiment, at the tilt compensate block **830** of the Kalman Filter, the processor receives an accelerometer data **220**. In some embodiments, the acceleration biases **810**

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may also be considered in calculating the tilt direction at the tilt compensate block **830**. Based on the tilt compensation, the processor outputs an acceleration vector that has a horizontal X, Y acceleration component. In some embodiments, a vertical Z acceleration component may also be calculated.

In one embodiment, at the correction block **840**, the processor receives misalignment angle and confidence information from the OTE block **740**, the motion status **240**, and gyroscope data **210**, along with the tilt compensated acceleration component from block **830** and forward acceleration, cross-track acceleration from block **820**. In other embodiments, the misalignment angle or the confidence value does not necessarily have to be received from an OTE block and could be based on other calculation of the angle such as using and SVD method or a GNSS DCM method or other suitable misalignment angle calculation method. While the acceleration bias block **810** and forward/cross-track acceleration block **820** are divided into separate blocks, these acceleration information can be retrieved from the accelerometer and, thus, from the accelerometer data **220**. In one embodiment, the correction block **840** receives the yaw rate from the gyroscope data **210**. Based on the above input, the processor computes the misalignment angle x at time t , x_t , and the error covariance P at time t , P_t . The misalignment angle x at time t , x_t , and the error covariance P at time t , P_t are output to the prediction block **850**. The prediction block **850** produces a misalignment angle $x_{t/t-1}$ at time $t-1$, and the error covariance $P_{t/t-1}$ at time $t-1$. After receiving the misalignment angle $x_{t/t-1}$ at time $t-1$ and the error covariance $P_{t/t-1}$ at time $t-1$, the processor computes a final misalignment angle and a final error covariance value and outputs at block **860**. The specific process within each block will be explained in detail in relation with FIG. 9.

FIG. 9 is a detailed flow diagram **900** of an exemplary Kalman Filter shown in FIG. 8 according to one embodiment of the present disclosure. A person of ordinary skill in the art will readily understand that the process set forth in the correction block **840** and the prediction block **850** using various filters is just one way of implementing the correction block **840** and the prediction block **850**. Other suitable filters and process may be added or altered or replaced based on the teachings of the present disclosure. In one embodiment, the correction block **840**, R, K, and H filters are included for the calculation of the X, Y acceleration components. Further, in the prediction block **850**, Q, A, Z filters are included for the calculation of the predictive X, Y acceleration components. In one embodiment, the uncertainty is represented by symbol "R" in the Kalman Filter.

Once the initial estimates for x_{k-1} and P_{k-1} is inputted to the prediction block, the Q, A, Z filters in the prediction block **850** processes the input. In one embodiment, the input projects the state ahead based on the following relationship:

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k$$

The prediction block **850** also projects the error covariance P_k^- ahead based on the following relationship:

$$P_k^- = AP_{k-1}A^T + Q$$

Once the prediction is made in the prediction block **850**, the input is provided to the correction block **840** including the R, K, and H filters.

In one embodiment, the Kalman gain K_k may be calculated based on the following formula.

$$K_k = P_k^- H^T (H P_k^- H^T + R)^{-1}$$

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The update estimate of x_k with measurement z_k can be expressed by the following formula:

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-)$$

The error covariance P_k may be expressed based on the following relationship:

$$P_k = (I - K_k H) P_k^-$$

After the angle based on the Kalman Filter and the error covariance is produced, the update of the prediction block and the update of the measurement from the correction block can occur recursively to produce a misalignment angle that produces the least amount of error and error covariance to the output block **860**.

Accordingly, the correction block **840** may update its calculation process by receiving the confidence from the OTE block **740**, motion status **240**, the yaw rate from the gyroscope data **210**, acceleration 2-dimensional norm error at the R filter block **842**. After the processing in the R filter block **842**, the output is inputted into the K filter block **844**. The K filter block **844** produces output so that the X, Y acceleration components can be inputted to the Z filter block **856**. At the Z filter block **856**, the covariance error P and the misalignment angle x are inputted. These series of processes may be performed based on the previously explained formula and variable relationship.

The measured input is then provided to the inverse Z transform block **856** of the prediction block **850**, and Z^{-1} outputs an angle $(t-1)$, an angle at time $t-1$ (angle x_{t-1}) and an error covariance P_{t-1} at time $t-1$. The output is received at the A filter block **854** along with the output from the Q filter block **852**. The misalignment angle $x_{t/t-1}$ at time $t-1$, and the error covariance $P_{t/t-1}$ at time $t-1$ are fed back to the correction block **840**. In particular, the misalignment angle $x_{t/t-1}$ is fed into the H filter block **846**, and based on the H filter block **846**, predicted X, Y acceleration components are produced. Similarly, this series of processes may be performed based on the previously explained formula and variable relationship as well.

These inputs are fed back into the correction block **840** and go through the previously explained process in a recursive manner to reduce and produce a final optimized misalignment angle and error covariance value at block **860**.

FIG. 10 is a flow diagram of a finalize estimation block **680** according to another embodiment of the present disclosure.

In one embodiment, the finalize estimation block **680** is responsible for when to override the existing misalignment angle. As explained in connection with FIG. 6, after the weight based merging block **670** produces a merged misalignment angle estimation, the angle is further examined to determine whether to produce this angle as a final estimate. In other embodiments, the finalize estimation block **680** may update the existing misalignment angle.

At the block **680**, a misalignment angle estimated by the Kalman Filters in the previous stage is provided along with the error covariance value P from the filters. The angle and the P value are inputted to the calculation validity metrics block **1010**. The calculation validity metrics block **1010** checks at block **1020** whether the error covariance value P is lower than the previous last P value. In some embodiments, at block **1020**, the error covariance value P is saved when it has the smallest value. Regarding the angle, the angle is fed into the First-in First-out (FIFO) block **1015** and the angle samples are outputted to the standard deviation calculation block **1025**. At block **1025**, the standard deviation is calculated for the angle samples and produces an angle to block **1050**.

When the validity process starts at block **1035**, the processor initially checks whether there is enough data or

samples accumulated for a validity check at block 1040. If there is not enough data, the processor acquires more data from the sensors at the next epoch 1055. If there is enough data, the processor proceeds to block 1045 to check if the current error covariance is lower than last P value. Here, block 1045 receives a P value based on the inputs from the calculation validity metrics block 1010, and it further checks if the current error covariance is lower than the last error covariance P value saved. If the error covariance value is not the lowest P value, the processor acquires more data from the sensors at the next epoch 1055. If the error covariance is the lowest P value, then the processor proceeds to block 1050 to check if the angle has the lowest standard deviation at block 1050. If the standard deviation of Kalman Filter misalignment angle is less than a given threshold, the process proceeds to the calculation final result block 1060. However, if the misalignment angle does not have the lowest standard deviation, then the processor acquires more data from the sensors at the next epoch 1055.

If the current Kalman Filter output, angle and error covariance all qualify in the previous step, the angle value is stored in a separate buffer 1065 of the calculation final result block 1060. The angle and error covariance values are only stored in the buffer if they meet the criteria in the validity check block 1030. Any angle or error covariance that fails to meet the criteria in the validity check block 1030 will be discarded. If the total sample of angles and error covariance values accumulated in the buffer are more than a given threshold value N and it is determined to be full at block 1065, the means and standard deviation is calculated at block 1080. In one or more embodiments, the threshold value N may be determined based on the level of precision. For example, as previously mentioned, if the final output of the misalignment angle is expected to have an accuracy that deviates less than 2 degrees from the actual angle, the threshold value may be calculated based on N number of samples, but if the accuracy is to be less than 1 degrees, the number of samples of the threshold value may be a number that is greater than N number of samples. If the buffer is not full, the buffer is updated at block 1070. The misalignment angle and the error covariance may be received from the next epoch 1075 to update the data. In some embodiments, selected angle samples from the updated buffer may be inputted to the mean and standard deviation calculation block 1080. The mean and standard deviation calculation block 1080 computes the standard deviation of misalignment angle value stored in the buffer. Based on the calculation of the mean and standard deviation, if the produced angle is stable then the block proceeds to block 1090 to conclude that the alignment is complete. Here, if the angle is stable, the standard deviation value should be small. When the calculation final result block 1060 concludes that the alignment is complete, it outputs the result at block 1095. In one embodiment, the output at the result block 1095 includes the current time, an average of the misalignment angle and the standard deviation of the misalignment angle stored in the buffer.

While various applications can be made based on the present disclosure, one exemplary application of the present disclosure according to one or more embodiments is the device 120 communicating the results of the dead reckoning to another electronic device. The device 120 may include modem chips, network and communication chips, or any suitable semiconductor chips capable of transmitting and receiving data. The device 120 may also communicate based on various communication protocols, including Wi-Fi™, Bluetooth™, Zigbee™, NFC™, cellular (e.g., 4G, 5G, LTE)

etc. In one application, the device 120 may relay the path information of the vehicle calculated based on the misalignment angle to a first device, for example, the vehicle driver's electronic device. Once the vehicle driver's electronic device receive the information, the same path information of the vehicle can be relayed to another user trying to locate the vehicle or the vehicle driver.

In a similar application, the device 120 may relay the path information of the vehicle calculated based on the misalignment angle to a second device, for example, a user's electronic device who wants an update location of the vehicle. The device 120 may directly send the path information of the vehicle to the user so that the user may obtain the position information of the vehicle.

In other applications, the device 120 may be incorporated to any devices within the vehicle. For example, the device 120 does not necessarily have to be a separate device and can be incorporated to any electronic devices within the vehicle such as navigation devices, vehicle black boxes, or other devices.

In further applications, the device 120 may be incorporated into the vehicle driver's electronic device. In yet further applications, the device 120 may be a standalone device that incorporates a chipset module, a receiver module or the like. In this application, the standalone device 120 can be installed in any orientation. The algorithm based on the present disclosure provides automatic correction of misalignment of the sensor of the device with respect to the reference frame of the vehicle. This provides great flexibility in that the device 120 does not have to be fixed in a specific orientation or at a specific location. That is, factory calibration to compensate or mitigate misalignment angle between the vehicle frame and the sensor frame is not necessary.

In one or more embodiments, the algorithm according to the present disclosure will operate reliably under GPS/GNSS signal absent environments. For example, a device 120 embodying the algorithm according to the present disclosure will be able to operate and track the path and position of the device 120 in less than 10 seconds after the GPS/GNSS signals become weak or lost.

In one or more embodiments, the algorithm according to the present disclosure will produce an accurate misalignment angle that has an error that deviates less than 2 degrees from the actual misalignment angle.

In other embodiments, the stand-alone device and related sensor data may be used to evaluate a collision between vehicles. The stand-alone device may be included in autonomous vehicles and human-operated vehicles, such as on the dash board or some other position selected by the driver or other party. This device is fixed with respect to the car during normal operation and will collect data before, during and after a collision. The accurate sensor alignment with respect to the vehicle reference frame assists in effectively analyzing crashes. For example, once a collision between vehicles has occurred, the aligned sensor data can be used to analyze the collision and to learn more about the circumstances surrounding the collision. The aligned sensor data is stored, such as in a memory in the device or in a remote server. Upon retrieval of this recorded data, the crash or collision can be analyzed. The misalignment angle between the sensor of the device and the vehicle reference frame and other data collected during the collision, is available even under circumstances where the GPS/GNSS signals are degraded or lost.

The various embodiments described above can be combined to provide further embodiments. All of the U.S. patents, U.S. patent application publications, U.S. patent

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applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

The invention claimed is:

1. A method comprising:

generating, by a gyroscope of a device, gyroscope data; generating, by an accelerometer of the device, accelerometer data;

generating, by a global positioning system sensor of the device, global positioning data;

determining a motion status of a vehicle based on the gyroscope data and the accelerometer data;

determining, using a first estimation method, a first estimated misalignment angle between a reference frame of the vehicle and a reference frame of the device based on the gyroscope data, the accelerometer data, and the motion status, the first estimation method including applying a singular value decomposition to the acceleration data;

determining, using a second estimation method, a second estimated misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the gyroscope data, the accelerometer data, the global positioning data, and the motion status, the second estimation method including applying an orthogonal transformation to the acceleration data and the global positioning data;

determining, using a third estimation method, a third estimated misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the gyroscope data, the accelerometer data, the global positioning data, and the motion status, the third estimation method including minimizing an error between the accelerometer data and predicted accelerometer data;

determining a misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the first, second, and third estimated misalignment angles; and

setting a location of the vehicle based on the misalignment angle.

2. The method of claim 1, further comprising:

determining a first weight for the first estimated misalignment angle;

determining a second weight for the second estimated misalignment angle;

determining a third weight for the third estimated misalignment angle; and

determining the misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the first, second, and third estimated misalignment angles and the first, second, and third weights.

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3. The method of claim 2, further comprising:

determining the first, second, and third estimated misalignment angles are within first, second, and third ranges, respectively; and

determining the first weight in response to determining the first estimated misalignment angle is within the first range;

determining the second weight in response to determining the second estimated misalignment angle is within the second range; and

determining the third weight in response to determining the third estimated misalignment angle is within the third range.

4. The method of claim 2 wherein the first, second, and third weights are determined based on a presence or absence of a sensor, a signal strength of the sensor, a motion characteristics of the vehicle, or a path condition of the vehicle.

5. The method of claim 1 wherein the third estimation method includes:

calculating the predicted accelerometer data based on the global positioning data.

6. The method of claim 1 wherein

the accelerometer data includes a first acceleration component in a first direction and a second acceleration component in a second direction transverse to the first direction, and

the global positioning data includes a first acceleration component in a third direction and a second acceleration component in a fourth direction transverse to the third direction.

7. The method of claim 1 wherein the motion status indicates whether the vehicle is moving in a straight line or making a turn.

8. The method of claim 1 wherein each of the first and second estimation methods includes:

determining a tilt direction of the device with respect to a horizontal plane; and

compensating the accelerometer data for the tilt direction.

9. The method of claim 1 wherein each of the first and second estimation methods includes:

filtering the gyroscope data and the accelerometer data for noise.

10. A device comprising:

a gyroscope configured to generate gyroscope data;

an accelerometer configured to generate accelerometer data;

a global positioning system sensor configured to generate global positioning data; and

a processor configured to:

determine a motion status of a vehicle based on the gyroscope data and the accelerometer data;

determine, using a first estimation method, a first estimated misalignment angle between a reference frame of the vehicle and a reference frame of the device based on the gyroscope data, the accelerometer data, and the motion status, the first estimation method including applying a singular value decomposition to the acceleration data;

determine, using a second estimation method, a second estimated misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the gyroscope data, the accelerometer data, the global positioning data, and the motion status, the second estimation method including applying an orthogonal transformation to the acceleration data and the global positioning data;

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determine, using a third estimation method, a third estimated misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the gyroscope data, the accelerometer data, the global positioning data, and the motion status, the third estimation method including minimizing an error between the accelerometer data and predicted accelerometer data;

determine a misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the first, second, and third estimated misalignment angles; and

setting a location of the vehicle based on the misalignment angle.

11. The device of claim 10 wherein the processor is configured to:

determine a first weight for the first estimated misalignment angle;

determine a second weight for the second estimated misalignment angle;

determine a third weight for the third estimated misalignment angle; and

determine the misalignment angle between the reference frame of the vehicle and the reference frame of the device based on the first, second, and third estimated misalignment angles and the first, second, and third weights.

12. The device of claim 11 wherein the processor is configured to:

determine the first, second, and third estimated misalignment angles are within first, second, and third ranges, respectively; and

determine the first weight in response to determining the first estimated misalignment angle is within the first range;

determine the second weight in response to determining the second estimated misalignment angle is within the second range; and

determine the third weight in response to determining the third estimated misalignment angle is within the third range.

13. The device of claim 11 wherein the first, second, and third weights are determined based on a presence or absence of a sensor, a signal strength of the sensor, a motion characteristics of the vehicle, or a path condition of the vehicle.

14. The device of claim 10 wherein the processor is configured to:

calculate the predicted accelerometer data based on the global positioning data.

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15. The device of claim 10 wherein the motion status indicates whether the vehicle is moving in a straight line or making a turn.

16. The device of claim 10 wherein the processor, for each of the first and second estimation methods, is configured to: determine a tilt direction of the device with respect to a horizontal plane; and

compensate the accelerometer data for the tilt direction.

17. The device of claim 10 wherein the processor, for each of the first and second estimation methods, is configured to: filter the gyroscope data and the accelerometer data for noise.

18. A method comprising:

receiving gyroscope data;

receiving accelerometer data;

receiving global positioning data;

determining, using a first estimation method, a first estimated misalignment angle between a first reference frame and a second reference frame based on the gyroscope data and the accelerometer data;

determining, using a second estimation method, a second estimated misalignment angle between the first reference frame and the second reference frame based on the gyroscope data, the accelerometer data, and the global positioning data;

determining, using a third estimation method, a third estimated misalignment angle between the first reference frame and the second reference frame based on the gyroscope data, the accelerometer data, and the global positioning data; and

determining a misalignment angle between the first reference frame and the second reference frame based on the first, second, and third estimated misalignment angles; and

setting a location of an object based on the misalignment angle,

the first estimation method includes applying a singular value decomposition to the acceleration data, the second estimation method includes applying an orthogonal transformation to the acceleration data and the global positioning data, and the third estimation method includes minimizing an error between the accelerometer data and predicted accelerometer data.

19. The method of claim 18, further comprising:

determining a motion status based on the gyroscope data and the accelerometer data, the first, second, and third estimated misalignment angles being determined based on the motion status.

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